

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	9
Warnings and Errors:	11
Input Echo:	12
XY Coordinate Calculations:	23
Internal Pressure Calculations:	24
Element: Bottom head	25
Element: Shell-1	27
Element: Shell-2	29
Element: Shell-2	31
Element: Top head	33
Shop Pressure Test Results:	35
Field Pressure Test Results:	37
External Pressure Calculations:	39
Element and Detail Weights:	40
Nozzle Flange MAWP:	43
Natural Frequency Calculation:	44
Forces/Moments Applied to Vessel	45
Wind Load Calculation:	46
Earthquake Load Calculation:	49
User Force/Moment Shear and Bendi	51
Wind/Earthquake Shear, Bending:	52
Wind/Earthquake Deflection:	53
Longitudinal Stress Constants:	54
Longitudinal Allowable Stresses:	55
Longitudinal Stresses due to:	56
Stress due to Combined Loads:	58
Skirt Stress Calculations:	63
Skirt Stress Calculations:	67
Center of Gravity Calculation:	72
Basering Calculations:	73
Nozzle Calcs.: N2_400	77
Nozzle Calcs.: U1_600	87
Nozzle Calcs.: K41_50	93
Nozzle Calcs.: K1_80	100
Nozzle Calcs.: K2_80	106
Nozzle Calcs.: K42_80	112
Nozzle Calcs.: M1_600	118
Nozzle Calcs.: K3_50	124
Nozzle Calcs.: K5_80	131
Nozzle Calcs.: N1_400	140



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\108-DB-DC-Rev 01.pvdb

Date of Analysis : Sep 9, 2026 5:18pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 4592.000 x 4300.000 mm ID	
Vessel Design Length, Tangent to Tangent	6700.00 mm
Distance of Bottom Tangent above Grade	4300.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	4000.00 mm
Internal Design Temperature	400 °C
Internal Design Pressure	6.411 MPa
External Design Temperature	400 °C
External Design Pressure	0.103 MPa
Maximum Allowable Working Pressure	6.43 MPa
External Max. Allowable Working Pressure	2.13 MPa
Shop Test Pressure	11.663 MPa
Field Test Pressure	11.062 MPa
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	10.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-105	...	...	K03504	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-36	...	...	K02600	
Base Bolting	SA-307 A	...	6 ≤ t ≤ 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	1.6000	N/A	Yes
Skirt-1	...	...	...	1.6000	N/A	Yes
Bottom head	6.411	0.10	6.430	3.0000	N/A	Yes
Shell-1	6.411	0.10	6.450	3.0000	N/A	Yes
Shell-2	6.411	0.10	6.450	3.0000	N/A	Yes
Shell-2	6.411	0.10	6.450	3.0000	N/A	Yes
Top head	6.411	0.10	6.430	3.0000	N/A	Yes

Element Types and Properties:

Element	"To" Elev	Element Length	Nominal Thickness	Finished Thickness	Reqd Thk Internal	Reqd Thk External	Long Eff	Circ Eff
---------	-----------	----------------	-------------------	--------------------	-------------------	-------------------	----------	----------

Type	mm	mm	mm	mm	mm	mm		
Skirt	-1000.0	3000.0	38.0	38.0	...	...	1.00	1.00
Skirt	0.0	1000.0	38.0	38.0	...	...	0.70	0.50
Sphere	0.0	0.0	80.0	73.0	72.8	10.2	1.00	1.00
Cylinder	3000.0	3000.0	146.0	146.0	145.1	23.3	1.00	1.00
Cylinder	6000.0	3000.0	146.0	146.0	145.0	23.3	1.00	1.00
Cylinder	6700.0	700.0	146.0	146.0	145.0	23.3	1.00	1.00
Sphere	6700.0	0.0	80.0	73.0	72.8	10.2	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2_400	WRC-107/537	0.991	n/a	Passed
N1_400	WRC-107/537	0.991	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	40234.9	558953	304829888	4.567E+09
20	30	3500	34085.1	553908	193304672	2.897E+09
30	40	3999.98	32035.2	550840	160232080	2.345E+09
40	50	5500	32035.1	530852	160231104	2.345E+09
50	60	8500	23460.6	388813	76953704	964670272
60	70	10350	14663.2	154073	19744790	150011600
70	80	10700	10654.9	70055.8	10880172	71536816

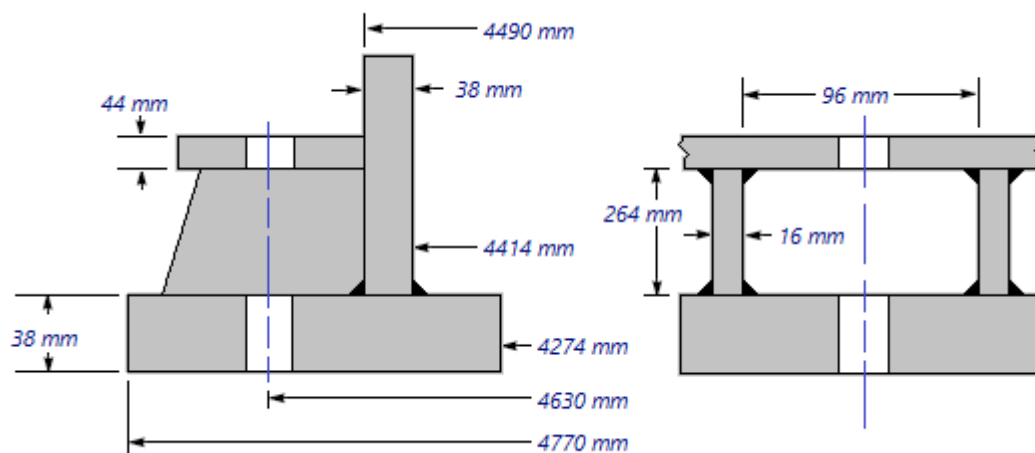
Abs Max of the all of the Stress Ratio's : 0.9758

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing 38.0000 mm
 Inside Diameter of Basing 4274.0000 mm
 Outside Diameter of Basing 4770.0000 mm
 Nominal Diameter of Bolts 48.0000 mm
 Diameter of Bolt Circle 4630.0000 mm
 Number of Bolts 24

Thickness of Gusset Plates 16.0000 mm
 Average Width of Gusset Plates 140.0000 mm
 Height of Gussets 264.0000 mm
 Distance between Gussets 96.0000 mm
 Thickness of Top Plate or Ring 44.0000 mm
 Radial Width of the Top Plate 140.0000 mm
 Number of Gussets 48

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	70570.	N
Total Earthquake Shear on Support	589288.	N
Wind Moment on Support	764438080.	N-mm
Earthquake Moment on Support	5026901504.	N-mm
Maximum Initial Bolt Load	38245.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	194210.7	kgm
Shop Test - Fabricated + water (full)	334963.8	kgm
Shipping - Fab. + rem. intls.+ shipping app.	352429.0	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	354828.0	kgm
Empty - Fab. + intls. + details + wghts.	354828.0	kgm
Operating - Empty + operating liquid (no ca)	363528.0	kgm
Field Test - Empty weight + test medium (full)	359241.4	kgm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2_400	6.43	OK	...	OK	17.24	OK	Passed
U1_600	6.43	OK	...	OK	21.57	OK	Passed
K41_50	6.43	OK	...	OK	7.80	OK	Passed
K41_50	6.43	...	...	OK	7.80	OK	No Calc[*]
K1_80	6.45	OK	...	OK	7.80	OK	Passed
K2_80	6.45	OK	...	OK	7.80	OK	Passed
K42_80	6.45	OK	...	OK	7.80	OK	Passed
M1_600	6.43	OK	...	...		OK	Passed
K3_50	6.43	OK	...	OK	7.80	OK	Passed
K3_50	6.43	...	...	OK	7.80	OK	No Calc[*]
K5_80	6.43	OK	...	OK	7.80	OK	Passed
N1_400	6.43	OK	...	OK	15.38	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6.43 Nozzle : N1400
 Minimum MAWP Shells/Flanges : 6.43 Element : Top head
 Minimum MAPnc Shells/Flanges : 9 Element : Shell-2

Computed Vessel M.A.W.P. : 6.43 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N2_400	0.000	0.000	739.151
30	U1_600	0.000	180.000	1124.000
30	K41_50	0.000	0.000	223.800
40	K1_80	2500.004	0.000	374.900
50	K2_80	5000.005	0.000	374.900
60	K42_80	6300.004	180.000	374.900
70	M1_600	0.000	0.000	1116.000
70	K3_50	0.000	0.000	224.400
70	K5_80	0.000	180.000	228.900
70	N1_400	0.000	60.000	748.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K41_50	50 mm	Actual	WNF	83.8	16.500	203.80	74.00	939	600
K3_50	50 mm	Actual	WNF	84.4	16.800	184.33	74.00	940	600
K1_80	80 mm	160	WNF	88.9	11.125	128.90	146.00	646	600
K2_80	80 mm	160	WNF	88.9	11.125	128.90	146.00	646	600
K42_80	80 mm	160	WNF	88.9	11.125	288.90	146.00	346	600
K5_80	80 mm	160	WNF	88.9	11.125	288.90	74.00	942	600
N2_400	400 mm	80	WNF	406.4	21.412	806.40	80.00	573	600
N1_400	400 mm	Actual	WNF	400.0	16.000	...	...	1299	600
U1_600	600 mm	Actual	WNF	600.0	22.000	840.00	146.00	2573	600
M1_600	600 mm	Actual	WNF	600.0	24.000	1200.00	80.00	493	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K41_50	SA-106 B	74.000	10.000	56.000	74.000	...
K3_50	SA-106 B	74.000	10.000	56.000	74.000	...
K1_80	SA-106 B	146.000	10.000	110.000	146.000	...
K2_80	SA-106 B	146.000	10.000	110.000	146.000	...
K42_80	SA-106 B	146.000	10.000	110.000	146.000	...
K5_80	SA-106 B	74.000	10.000	56.000	74.000	...
N2_400	SA-106 B	73.000	10.000	80.000	80.000	...
N1_400	SA-105	73.000	10.000	...	...	...
U1_600	SA-516 70	74.000	10.000	110.000	146.000	...
M1_600	SA-516 70	73.000	10.000	56.000	80.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K41_50	...	0.0	800.00	0.00	Bottom head
K3_50	...	0.0	800.00	0.00	Top head
K1_80	2500.004	0.0	300.00	200.00	Shell-1
K2_80	5000.004	0.0	300.00	200.00	Shell-2
K42_80	6300.003	180.0	200.00	0.00	Shell-2
K5_80	...	180.0	800.00	0.00	Top head
N2_400	...	0.0	200.00	300.00	Bottom head
N1_400	...	60.0	700.00	500.00	Top head
U1_600	...	180.0	1500.00	1000.00	Bottom head
M1_600	...	0.0	400.00	0.00	Top head



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 80.0mm THK X 4364.0mm ID X 0.0mm	SA-516 70
2	INSULATION: 2179mm X 150mm THK	
3	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 4300.0mm	...
2	PACKING: 3000.000mm	...
2	INSULATION: 3000mm X 150mm THK	
1	LADDER: 4000.0mm	...
1	LADDER: 3000.0mm	...
1	PACKING: 700.000mm	...
1	INSULATION: 700mm X 150mm THK	
1	PLATFORM: 1000mm X 5000mm WIDE	
1	LADDER: 5000.0mm	...
2	CLASS 600 GR 1.1, 600.0mm BLIND FLANGE(S)	SA-105
1	CLASS 600 GR 1.1, 50.8mm BLIND FLANGE(S)	SA-105
4	CLASS 600 GR 1.1, 80.0mm BLIND FLANGE(S)	SA-105
1	BASERING: 4770.0mm OD X 4274.0 ID X 38.0mm	SA-36
24	BASERING BOLTS: 48.0mm DIA	SA-307 A
24	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 4770.0mm OD X 4490.0 ID X 44.0mm	...
48	BASERING GUSSET PLATES: 140.0mm X 140.0mm X 264.0mm X 16.0mm	...
3	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3081.5mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3120.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 1502.2mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3081.5mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3120.0mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 1502.2mm X 1000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 146.0mm THK X 5365.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 146.0mm THK X 2776.8mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 146.0mm THK X 3039.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 146.0mm THK X 3077.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 146.0mm THK X 10469.2mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 146.0mm THK X 6116.4mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 146.0mm THK X 3039.4mm X 700.0mm	SA-516 70



1	CYLINDRICAL SEGMENT 146.0mm THK X 3077.0mm X 700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 146.0mm THK X 10469.2mm X 700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 146.0mm THK X 6116.4mm X 700.0mm	SA-516 70
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Bottom head	[9]	B	25	10		0.731	73.000	1.00	Yes
Shell-1	[17]	Impact testing required							
Shell-2	[17]	Impact testing required							
Shell-2	[17]	Impact testing required							
Top head	[9]	B	25	10		0.731	73.000	1.00	Yes
N2400	[1]	B	25	10		0.728	73.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
U1600	[1]	B	25	10		0.728	73.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K4150	[1]	B	25	10		0.728	73.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K180	[1]	B	-8	-104		0.303	9.735	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K280	[1]	B	-8	-104		0.303	9.735	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K4280	[1]	B	-8	-48		0.351	9.735	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
M1600	[1]	B	25	10		0.728	73.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K350	[1]	B	25	10		0.728	73.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K580	[1]	B	25	10		0.728	73.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
N1400	[1]	B	25	10		0.728	73.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No
Warmest MDMT:			25	10					
Required Minimum Design Metal Temperature						-12.0	°C		
Warmest Computed Minimum Design Metal Temperature						10.0	°C		

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:



Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Class From To : Basic Element Checks.

Note	10	30	There is a high jump in the Joint Eff.
Note	20	40	There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	6.411	MPa
Design Internal Temperature	400.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10



Element To Node	20
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	4490 mm
Diameter of Skirt at Base	4490 mm
Skirt Thickness	38 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	400 °C
Design Temperature External Pressure	50 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	101.03 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	181 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Weight
Detail ID	Tailing lug
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	14709 N
Offset from Element Centerline	0 mm

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	1000 mm
Skirt Outside Diameter	4490 mm
Diameter of Skirt at Base	4490 mm
Skirt Thickness	38 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	400 °C
Design Temperature External Pressure	400 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Spherical
Description	Bottom head
Distance "FROM" to "TO"	0 mm
Inside Diameter	4364 mm
Element Thickness	73 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	80 mm
External Corrosion Allowance	0 mm



Design Internal Pressure	6.411	MPa
Design Temperature Internal Pressure	400	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	400	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-2179	mm
Height/Length of Insulation	2179	mm
Thickness of Insulation	150	mm
Density	0.000352	kg/cm³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2400	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	400	mm
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	U1600	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	14833	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K4150	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Weight	
Detail ID	Closure plate	
Dist. from "FROM" Node / Offset dist	-200	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder



Description	Shell-1	
Distance "FROM" to "TO"	3000	mm
Element Outside Diameter	4592	mm
Element Thickness	146	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	146	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	6.411	MPa
Design Temperature Internal Pressure	400	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	400	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Platform	
Detail ID	PL-1	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	2999999	mm ²
Platform Weight	52380	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4300	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm
Element From Node	40	
Detail Type	Packing	
Detail ID	Catalyst	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	3000	mm
Density	0.001375	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	150	mm
Density	0.000352	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K180	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	80	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Weight	
Detail ID	Hold grid	



Dist. from "FROM" Node / Offset dist	200	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	0	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	SF wt2	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	33831	N
Offset from Element Centerline	0	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	ball	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	26476	N
Offset from Element Centerline	0	mm

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4300	mm
Element Thickness	146	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	146	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	6.411	MPa
Design Temperature Internal Pressure	400	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	400	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Platform	
Detail ID	PL-2	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	2999999	mm ²
Platform Weight	26875	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm

Element From Node	50	
Detail Type	Packing	
Detail ID	Catalyst	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	3000	mm
Density	0.001375	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	



Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	150 mm
Density	0.000352 kg/cm³

Element From Node	50
Detail Type	Nozzle
Detail ID	K280
Dist. from "FROM" Node / Offset dist	2000 mm
Nozzle Diameter	80 mm
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	50
Detail Type	Weight
Detail ID	Clip wt
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	32066 N
Offset from Element Centerline	0 mm

Element From Node	50
Detail Type	Weight
Detail ID	Trunnion 1
Dist. from "FROM" Node / Offset dist	2700 mm
Miscellaneous Weight	19612 N
Offset from Element Centerline	2500 mm

Element From Node	50
Detail Type	Weight
Detail ID	Trunnion-2
Dist. from "FROM" Node / Offset dist	2700 mm
Miscellaneous Weight	19612 N
Offset from Element Centerline	2500 mm

Element From Node	50
Detail Type	Weight
Detail ID	misc margin
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	98060 N
Offset from Element Centerline	0 mm

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	700 mm
Inside Diameter	4300 mm
Element Thickness	146 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	146 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6.411 MPa
Design Temperature Internal Pressure	400 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	400 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0



Weld is pre-Heated

No

Element From Node	60
Detail Type	Platform
Detail ID	PL-3
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	2999999 mm ²
Platform Weight	26505 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	185 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	3000 mm
Unit Weight of Ladder	0.37 N/mm
Platform Length (top head platform)	0 mm

Element From Node	60
Detail Type	Packing
Detail ID	Catalyst
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	700 mm
Density	0.001375 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	700 mm
Thickness of Insulation	150 mm
Density	0.000352 kg/cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	K4280
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	80 mm
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
Detail Type	Weight
Detail ID	Hold down plate
Dist. from "FROM" Node / Offset dist	400 mm
Miscellaneous Weight	980.6 N
Offset from Element Centerline	0 mm

Element From Node	60
Detail Type	Weight
Detail ID	Hold down plate
Dist. from "FROM" Node / Offset dist	300 mm
Miscellaneous Weight	980.6 N
Offset from Element Centerline	300 mm

Element From Node	60
Detail Type	Weight
Detail ID	Balls weight
Dist. from "FROM" Node / Offset dist	200 mm



Miscellaneous Weight 58836 N
 Offset from Element Centerline 100 mm

Element From Node 60
 Detail Type Weight
 Detail ID Hold down plate
 Dist. from "FROM" Node / Offset dist 0.004651 mm
 Miscellaneous Weight 980.6 N
 Offset from Element Centerline 0 mm

Element From Node 60
 Detail Type Weight
 Detail ID SF wt-1
 Dist. from "FROM" Node / Offset dist 0.004651 mm
 Miscellaneous Weight 33831 N
 Offset from Element Centerline 0 mm

Element# 7/7
 Element From Node 70
 Element To Node 80
 Element Type Spherical
 Description Top head
 Distance "FROM" to "TO" 0 mm
 Inside Diameter 4364 mm
 Element Thickness 73 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 80 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 6.411 MPa
 Design Temperature Internal Pressure 400 °C
 Design External Pressure 0.1034 MPa
 Design Temperature External Pressure 400 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 70
 Detail Type Platform
 Detail ID PL-4
 Dist. from "FROM" Node / Offset dist 2200 mm
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 0.0
 Platform Wind Area 703200 mm²
 Platform Weight 43547 N
 Platform Railing Weight 0 N/mm
 Platform Grating Weight 0.017 kg/cm²
 Platform Width 5000 mm
 Platform Height 1000 mm
 Platform Clearance or End Offset 2500 mm
 Platform Force Coefficient 1.0
 Ladder Layout Angle 90.0
 Ladder Start Elevation 0 mm
 Ladder End Elevation 5000 mm
 Unit Weight of Ladder 0.37 N/mm
 Platform Length (top head platform) 5000 mm

Element From Node 70
 Detail Type Insulation
 Detail ID Ins: 50
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 2179 mm
 Thickness of Insulation 150 mm
 Density 0.000352 kg/cm³

Element From Node 70
 Detail Type Nozzle



Detail ID M1600
 Dist. from "FROM" Node / Offset dist 0 mm
 Nozzle Diameter 600 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70 +IT

Element From Node 70
 Detail Type Nozzle
 Detail ID K350
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 50.8 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 70
 Detail Type Nozzle
 Detail ID K580
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 80 mm
 Nozzle Schedule 160
 Nozzle Class 600
 Layout Angle 180.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B +IT

Element From Node 70
 Detail Type Nozzle
 Detail ID N1400
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 400 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 60.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105 +IT

Element From Node 70
 Detail Type Weight
 Detail ID N1 wt
 Dist. from "FROM" Node / Offset dist 0 mm
 Miscellaneous Weight 16660 N
 Offset from Element Centerline 3748 mm

Element From Node 70
 Detail Type Weight
 Detail ID N1vane distrib
 Dist. from "FROM" Node / Offset dist 0 mm
 Miscellaneous Weight 2941.8 N
 Offset from Element Centerline 0 mm

Element From Node 70
 Detail Type Weight
 Detail ID M1davit
 Dist. from "FROM" Node / Offset dist 0 mm
 Miscellaneous Weight 980.6 N
 Offset from Element Centerline 0 mm



Element From Node	70
Detail Type	For./Mom.
Detail ID	N1 F&M
Dist. from "FROM" Node / Offset dist	2182 mm
Force in X Direction	21450 N
Force in Y Direction	29700 N
Force in Z Direction	21450 N
Moment about X Axis	19800 N-mm
Moment about Y Axis	0 N-mm
Moment about Z Axis	19800 N-mm
Force/Moment Combination Method	SRSS



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	Skirt-2	...	3000	...	3000
	Skirt-1	...	4000	...	1000
	Bottom head	...	4000	...	...
	Shell-1	...	7000	...	3000
	Shell-2	...	10000	...	3000
	Shell-2	...	10700	...	700
	Top head	...	10700	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2		...	38	1.6	4490	...	1
Skirt-1		...	38	1.6	4490	...	2
Bottom head		6.411	80	3	4364	101.03	3
Shell-1		6.411	146	3	4592	101.03	4
Shell-2		6.411	146	3	4300	101.03	5
Shell-2		6.411	146	3	4300	101.03	6
Top head		6.411	80	3	4364	101.03	7

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2		...	No Calc	No Calc	38	No Calc	1
Skirt-1		...	No Calc	No Calc	38	No Calc	2
Bottom head		6.411	6.43	9.17	73	72.7705	3
Shell-1		6.411	6.45	9	146	145.093	4
Shell-2		6.411	6.45	9	146	145.033	5
Shell-2		6.411	6.45	9	146	145.033	6
Top head		6.411	6.43	9.17	73	72.7705	7

Minimum 6.43 9

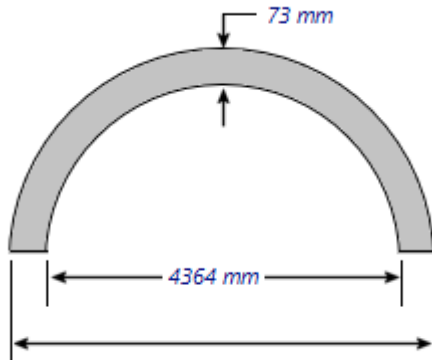
MAWP: 6.43 MPa, limited by: Top head.

Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom head

Spherical Head Design Information:					
Design Pressure	P	6.411 MPa	Int. Design Temperature		400 °C
Material		SA-516 70	Ext. Design Temperature		400 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	101.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4364.000 mm	Actual Stress		100.7 MPa
Spec. Min. Thk t,e, or sa		73.0000 mm	Spec. Nominal Thk.		80.000 mm
Surface Area		32148820.0 mm²	SG of Contents		0.000
Element Volume		21762.1 ltr	Weight of Contents		0.0 N
Empty Weight		188636.8 N	Operating Weight		188636.8 N
			Comp. All. Ope. Stress		82.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 54.550 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 27.275 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (6.41 \cdot 2185) / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$$

$$= 69.7705 + 3.0000 = 72.7705 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 101 \cdot 1 \cdot 70) / (2185 + 0.2 \cdot 70)$$

$$= 6.43 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 73) / (2182 + 0.2 \cdot 73)$$

$$= 9.17 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6.41 \cdot (2185 + 0.2 \cdot 70)) / (2 \cdot 1 \cdot 70)$$

= 100.698 MPa

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.700 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 73$, $t_r = 51.2$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.73$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 25 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 10 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.432 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	73.000 mm	Basic MDMT	25 °C
Stress/Thk. Ratio	0.731	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	10 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 54.550 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 27.275 mm

EXTERNAL PRESSURE RESULTS for: Bottom head

Spherical Head From 30 to 40 Ext. Chart: CS-2 at 400 °C

Elastic Modulus from Chart: CS-2 at 400 °C: 162839 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
70.000	4510.00	64.43	0.0038803	68.72

$$MAEP = B / ((Do/t)/2) = 68.7 / (64.4/2) = 2.13 \text{ MPa}$$

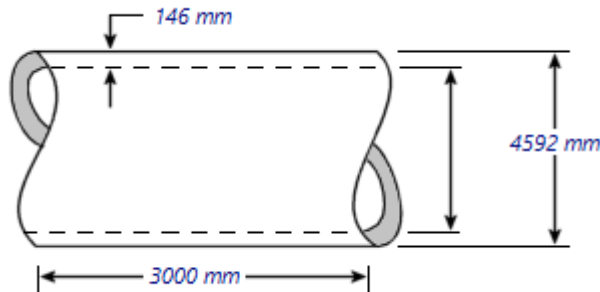
Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
7.189	4510.00	627.37	0.0003985	32.44

$$MAEP = B / ((Do/t)/2) = 32.4 / (627/2) = 0.1 \text{ MPa}$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	6.411 MPa	Int. Design Temperature		400 °C
Material		SA-516 70	Ext. Design Temperature		400 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	101.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Outside Diameter	Do	4592.000 mm	Actual Stress		100.4 MPa
Spec. Min. Thk t _e , or s _a		146.0000 mm	Spec. Nominal Thk.		146.000 mm
Surface Area		43278580.0 mm ²	SG of Contents		0.000
Element Volume		43573.8 ltr	Weight of Contents		0.0 N
Empty Weight		464959.2 N	Operating Weight		464959.2 N
			Comp. All. Ope. Stress		92.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (6.41 \cdot 2296) / (101 \cdot 1 + 0.4 \cdot 6.41)$$

$$= 142.0926 + 3.0000 = 145.0926 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (101 \cdot 1 \cdot 143) / (2296 - 0.4 \cdot 143)$$

$$= 6.45 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138 \cdot 1 \cdot 146) / (2296 - 0.4 \cdot 146)$$

$$= 9 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (6.41 \cdot (2296 - 0.4 \cdot 143)) / (1 \cdot 143)$$

$$= 100.370 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 3.284 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the



governing thickness for this component was greater than 100.000 mm.
This component must be impact tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.432 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	146.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 400 °C

Elastic Modulus from Chart: CS-2 at 400 °C: 162839 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
143.000	4592.00	8154.67	32.11	1.7758	0.0040461	69.21

$$MAEP = (4*B)/(3*(Do/t)) = (4*69.2)/(3*32.1) = 2.87 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.262	4592.00	8154.67	226.63	1.7758	0.0002159	17.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.6)/(3*227) = 0.1 \text{ MPa}$$

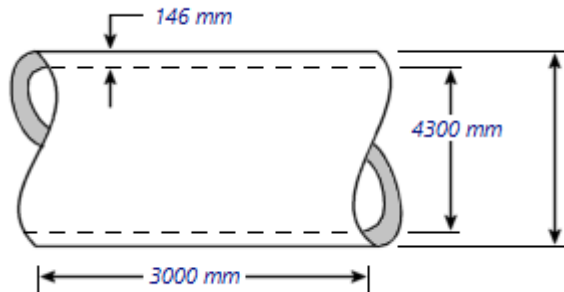
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
143.000	4592.00	199124.77	32.11	43.3634	0.0010865	53.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*53)/(3*32.1) = 2.2 \text{ MPa}$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	6.411 MPa	Int. Design Temperature		400 °C
Material		SA-516 70	Ext. Design Temperature		400 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	101.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4300.000 mm	Actual Stress		100.4 MPa
Spec. Min. Thk t _e , or s _a		146.0000 mm	Spec. Nominal Thk.		146.000 mm
Surface Area		43278580.0 mm ²	SG of Contents		0.000
Element Volume		43573.8 ltr	Weight of Contents		0.0 N
Empty Weight		464959.2 N	Operating Weight		464959.2 N
			Comp. All. Ope. Stress		92.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6.41 \cdot 2153) / (101 \cdot 1 - 0.6 \cdot 6.41)$$

$$= 142.0327 + 3.0000 = 145.0327 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (101 \cdot 1 \cdot 143) / (2153 + 0.6 \cdot 143)$$

$$= 6.45 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 146) / (2150 + 0.6 \cdot 146)$$

$$= 9 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (6.41 \cdot (2153 + 0.6 \cdot 143)) / (1 \cdot 143)$$

$$= 100.370 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 3.284 %

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the

governing thickness for this component was greater than 100.000 mm.
 This component must be impact tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.432 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	146.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 400 °C

Elastic Modulus from Chart: CS-2 at 400 °C: 162839 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
143.000	4592.00	8154.67	32.11	1.7758	0.0040461	69.21

$$MAEP = (4*B)/(3*(Do/t)) = (4*69.2)/(3*32.1) = 2.87 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.262	4592.00	8154.67	226.63	1.7758	0.0002159	17.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.6)/(3*227) = 0.1 \text{ MPa}$$

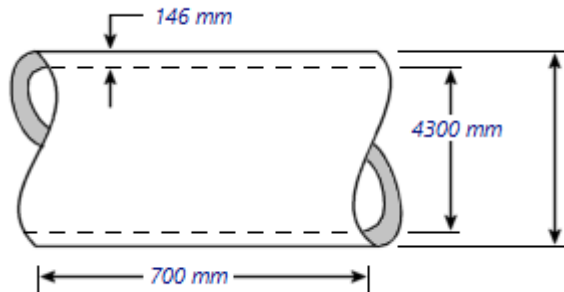
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
143.000	4592.00	199124.77	32.11	43.3634	0.0010865	53.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*53)/(3*32.1) = 2.2 \text{ MPa}$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	6.411 MPa	Int. Design Temperature		400 °C
Material		SA-516 70	Ext. Design Temperature		400 °C
Length		700.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	101.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4300.000 mm	Actual Stress		100.4 MPa
Spec. Min. Thk t _e , or s _a		146.0000 mm	Spec. Nominal Thk.		146.000 mm
Surface Area		10098336.0 mm ²	SG of Contents		0.000
Element Volume		10167.2 ltr	Weight of Contents		0.0 N
Empty Weight		108490.5 N	Operating Weight		108490.5 N
			Comp. All. Ope. Stress		92.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6.41 \cdot 2153) / (101 \cdot 1 - 0.6 \cdot 6.41)$$

$$= 142.0327 + 3.0000 = 145.0327 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (101 \cdot 1 \cdot 143) / (2153 + 0.6 \cdot 143)$$

$$= 6.45 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 146) / (2150 + 0.6 \cdot 146)$$

$$= 9 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (6.41 \cdot (2153 + 0.6 \cdot 143)) / (1 \cdot 143)$$

$$= 100.370 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 3.284 %

Minimum Design Metal Temperature Results:

Warning:
 The minimum metal temperature could not be computed because the

governing thickness for this component was greater than 100.000 mm.
 This component must be impact tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.432 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	146.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 400 °C

Elastic Modulus from Chart: CS-2 at 400 °C: 162839 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
143.000	4592.00	8154.67	32.11	1.7758	0.0040461	69.21

$$MAEP = (4*B)/(3*(Do/t)) = (4*69.2)/(3*32.1) = 2.87 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.262	4592.00	8154.67	226.63	1.7758	0.0002159	17.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.6)/(3*227) = 0.1 \text{ MPa}$$

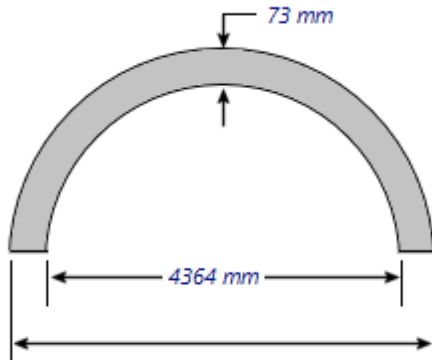
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
143.000	4592.00	199124.77	32.11	43.3634	0.0010865	53.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*53)/(3*32.1) = 2.2 \text{ MPa}$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top head

Spherical Head Design Information:					
Design Pressure	P	6.411 MPa	Int. Design Temperature		400 °C
Material		SA-516 70	Ext. Design Temperature		400 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	101.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	4364.000 mm	Actual Stress		100.7 MPa
Spec. Min. Thk t,e, or sa		73.0000 mm	Spec. Nominal Thk.		80.000 mm
Surface Area		32148820.0 mm²	SG of Contents		0.000
Element Volume		21762.1 ltr	Weight of Contents		0.0 N
Empty Weight		188636.8 N	Operating Weight		188636.8 N
			Comp. All. Ope. Stress		82.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 54.550 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 27.275 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (6.41 \cdot 2185) / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$$

$$= 69.7705 + 3.0000 = 72.7705 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 101 \cdot 1 \cdot 70) / (2185 + 0.2 \cdot 70)$$

$$= 6.43 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 73) / (2182 + 0.2 \cdot 73)$$

$$= 9.17 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6.41 \cdot (2185 + 0.2 \cdot 70)) / (2 \cdot 1 \cdot 70)$$

= 100.698 MPa

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.700 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 73$, $t_r = 51.2$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.731$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 25 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 10 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.432 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	73.000 mm	Basic MDMT	25 °C
Stress/Thk. Ratio	0.731	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	10 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 54.550 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 27.275 mm

EXTERNAL PRESSURE RESULTS for: Top head

Spherical Head From 70 to 80 Ext. Chart: CS-2 at 400 °C

Elastic Modulus from Chart: CS-2 at 400 °C: 162839 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
70.000	4510.00	64.43	0.0038803	68.72

$$MAEP = B / ((Do/t)/2) = 68.7 / (64.4/2) = 2.13 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
7.189	4510.00	627.37	0.0003985	32.44

$$MAEP = B / ((Do/t)/2) = 32.4 / (627/2) = 0.1 \text{ MPa}$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	11.099 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	11.062 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	11.663 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	9.391 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	10.637 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	11.062 MPa
App 27-4	= M.A.W.P.	6.432 MPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	176.1	234.0	0.753	11.71
Shell-1	179.4	234.0	0.767	11.71
Shell-2	179.4	234.0	0.767	11.70
Shell-2	179.4	234.0	0.767	11.70
Top head	176.1	234.0	0.753	11.71

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2_400	Nozzle	118.00	88.90	1.327
N2_400	Pad	138.00	101.00	1.366
U1_600	Nozzle	138.00	101.00	1.366
U1_600	Pad	138.00	101.00	1.366
K41_50	Nozzle	118.00	88.90	1.327
K41_50	Pad	138.00	101.00	1.366
K1_80	Nozzle	118.00	88.90	1.327
K1_80	Pad	138.00	101.00	1.366
K2_80	Nozzle	118.00	88.90	1.327
K2_80	Pad	138.00	101.00	1.366
K42_80	Nozzle	118.00	88.90	1.327
K42_80	Pad	138.00	101.00	1.366
M1_600	Nozzle	138.00	101.00	1.366
M1_600	Pad	138.00	101.00	1.366
K3_50	Nozzle	118.00	88.90	1.327
K3_50	Pad	138.00	101.00	1.366
K5_80	Nozzle	118.00	88.90	1.327
K5_80	Pad	138.00	101.00	1.366
N1_400	Nozzle	138.00	101.00	1.366

Minimum 1.327

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	101.03	1.366
Shell-1	138.00	101.03	1.366
Shell-2	138.00	101.03	1.366
Shell-2	138.00	101.03	1.366
Top head	138.00	101.03	1.366

Minimum 1.366

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



N2_400	106.40	216.00	0.493
U1_600	154.94	179.40	0.864
K41_50	25.04	216.00	0.116
K1_80	42.09	216.00	0.195
K2_80	42.08	216.00	0.195
K42_80	42.08	216.00	0.195
M1_600	141.63	179.40	0.789
K3_50	24.72	216.00	0.114
K5_80	48.77	216.00	0.226
N1_400	141.63	225.00	0.629

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	11.099 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	11.062 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	11.677 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	9.391 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	10.637 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	11.062 MPa
App 27-4	= M.A.W.P.	6.432 MPa

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 6.41 • 1.33
 = 11.062 MPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	175.5	234.0	0.750	11.17
Shell-1	174.6	234.0	0.746	11.15
Shell-2	174.1	234.0	0.744	11.12
Shell-2	173.6	234.0	0.742	11.09
Top head	174.1	234.0	0.744	11.08

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2_400	Nozzle	118.00	88.90	1.327
N2_400	Pad	138.00	101.00	1.366
U1_600	Nozzle	138.00	101.00	1.366
U1_600	Pad	138.00	101.00	1.366
K41_50	Nozzle	118.00	88.90	1.327
K41_50	Pad	138.00	101.00	1.366
K1_80	Nozzle	118.00	88.90	1.327
K1_80	Pad	138.00	101.00	1.366
K2_80	Nozzle	118.00	88.90	1.327
K2_80	Pad	138.00	101.00	1.366
K42_80	Nozzle	118.00	88.90	1.327
K42_80	Pad	138.00	101.00	1.366
M1_600	Nozzle	138.00	101.00	1.366
M1_600	Pad	138.00	101.00	1.366
K3_50	Nozzle	118.00	88.90	1.327
K3_50	Pad	138.00	101.00	1.366
K5_80	Nozzle	118.00	88.90	1.327
K5_80	Pad	138.00	101.00	1.366
N1_400	Nozzle	138.00	101.00	1.366

Minimum 1.327

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	101.03	1.366
Shell-1	138.00	101.03	1.366
Shell-2	138.00	101.03	1.366
Shell-2	138.00	101.03	1.366
Top head	138.00	101.03	1.366

Minimum 1.366



Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N2_400	101.54	216.00	0.470
U1_600	147.86	179.40	0.824
K41_50	23.90	216.00	0.111
K1_80	40.09	216.00	0.186
K2_80	39.98	216.00	0.185
K42_80	39.88	216.00	0.185
M1_600	134.11	179.40	0.748
K3_50	23.41	216.00	0.108
K5_80	46.18	216.00	0.214
N1_400	134.11	225.00	0.596

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	4000	...	...	No Calc	No Calc
20	30	4000	...	...	No Calc	No Calc
30	40	No Calc	4510	70	0.0038803	68.72
40	50	8154.67	4592	143	0.0040461	69.2149
50	60	8154.67	4592	143	0.0040461	69.2149
60	70	8154.67	4592	143	0.0040461	69.2149
70	80	No Calc	4510	70	0.0038803	68.72

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	73	10.1887	0.10343	2.13
40	50	146	23.2623	0.10343	2.87
50	60	146	23.2623	0.10343	2.87
60	70	146	23.2623	0.10343	2.87
70	80	73	10.1887	0.10343	2.13

Minimum

2.13

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	4000	No Calc	No Calc	No Calc
20	30	4000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	8154.67	199125	No Calc	No Calc
50	60	8154.67	199125	No Calc	No Calc
60	70	8154.67	199125	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	14309.8	...	13789.6	...	...
20	30	4119.25	...	3945.82	...	...
30	40	19236.9	21762.1	18540.3	21852	...
40	50	47415.8	43573.8	46472.9	43695.5	...
50	60	47415.8	43573.8	46472.9	43695.5	...
60	70	11063.7	10167.2	10843.7	10195.6	...
70	80	19236.9	21762.1	18540.3	21852	...
Total		162798	140839.11	158605	141290.62	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	1500	...	...	...	Tailing lug
30	Insl	1887.65	...	-1089.5	...	Ins: 10
30	Nozl	539.12	...	-2095.5	...	N2_400
30	Nozl	1512.6	-1500	-1798.15	...	U1_600
30	Nozl	111.36	1500	-3113.12	...	K41_50
30	Wght	200	...	-200	...	Closure plate
40	Plat	5341.66	0.27855E-04	...	...	PL-1
40	Pack	59973.4	...	1500	...	Catalyst
40	Insl	2362.52	...	1500	...	Ins: 20
40	Nozl	39.1795	2300	2500	...	K1_80
40	Wght	100	...	200	...	Hold grid
40	Wght	3450	...	...	...	SF wt2
40	Wght	2700	...	...	...	ball
50	Plat	2740.63	...	...	-2002.66	PL-2
50	Pack	59973.4	...	1500	...	Catalyst
50	Insl	2362.52	...	1500	...	Ins: 30
50	Nozl	39.1795	2300	2000	...	K2_80
50	Wght	3270	...	1500	...	Clip wt
50	Wght	2000	2500	2700	...	Trunnion 1
50	Wght	2000	-2500	2699.99	...	Trunnion-2
50	Wght	10000	...	...	...	misc margin
60	Plat	2702.9	...	...	-2002.66	PL-3
60	Pack	13993.8	...	350	...	Catalyst
60	Insl	551.254	...	350	...	Ins: 40
60	Nozl	92.2665	-2250	300.004	...	K42_80
60	Wght	100	...	400.005	...	Hold down plate
60	Wght	100	300	300.004	...	Hold down plate
60	Wght	6000	100	200.005	...	Balls weight
60	Wght	100	...	0.0046512	...	Hold down plate
60	Wght	3450	...	0.0046512	...	SF wt-1
70	Plat	4440.85	...	2200	...	PL-4
70	Insl	1887.65	...	1089.5	...	Ins: 50
70	Nozl	1642.26	...	2345.5	...	M1_600
70	Nozl	41.2386	1500	1948.15	...	K3_50
70	Nozl	68.2269	-1500	1948.15	...	K5_80
70	Nozl	1357.09	750	1648.15	-1299.04	N1_400
70	Wght	1699	3748	...	...	N1 wt
70	Wght	300	...	...	...	N1_Vane distrib
70	Wght	100	...	...	...	M1_Davit
70	Forc	...	...	2182	...	N1 F&M

Total Weight of Each Detail Type:

Platforms	15226.0
Packing	133940.7
Insulation	9051.6



Nozzles 5442.5
Weights 37069.0

Sum of the Detail Weights 200729.8 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	162798.1	162798.1	162798.1	162798.1	162798.1	162798.1
Nozzles	5442.5	5442.5	5442.5	5442.5	5442.5	5442.5
Wld Weights	25970.0	25970.0	25970.0	25970.0	25970.0	25970.0
Platforms	...	...	15226.0	15226.0	15226.0	15226.0
Packing	...	...	133940.7	133940.7	133940.7	133940.7
Insulation	...	...	9051.6	9051.6	9051.6	9051.6
Empty Weights	...	...	...	2399.0	2399.0	...
Ope Weights	...	...	...	...	...	11099.0
Test Liquid	...	140753.1	...	...	...	...
Totals	194210.7	334963.8	352429.0	354828.0	354828.0	363528.0

Field Installation Options:

Weight Summary:

Fabricated Wt. - Bare weight without removable internals 194210.7 kgm
 Shop Test Wt. - Fabricated weight + water (full) 334963.8 kgm
 Shipping Wt. - Fab. weight + removable intls.+ shipping app. 352429.0 kgm
 Erected Wt. - Fab. wt + or - loose items (trays,platforms etc.) 354828.0 kgm
 Ope. Wt. no Liq - Fab. weight + internals. + details + weights 354828.0 kgm
 Operating Wt. - Empty weight + operating liq. uncorroded 363528.0 kgm
 Field Test Wt. - Empty weight + test medium (full) 359241.4 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating) 180916.5 kgm
 Mass of the Upper 1/3 of the Vertical Vessel (filled w/water) 202762.2 kgm
 Mass of the Upper 1/3 of the Vertical Vessel (empty) 176734.1 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	42317252	41600972
20	30	14105751	13866990
30	40	32148820	29915024
40	50	43278580	40526544
50	60	43278580	40526544
60	70	10098336	9456194
70	80	32148820	29915024

Total 217376144.0 205807280.0 mm²
 Total 217.4 205.8 m²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	15809.8	15809.8	15809.8	...	15809.8
20	30	4119.25	4119.25	4119.25	...	4119.25
30	40	23487.6	23487.6	45236.4	...	23487.6
40	50	118683	121383	102156	...	121383
50	60	129802	129802	113375	53842628	129802
60	70	32153.9	38153.9	28021.1	53101340	38153.9
70	80	30773.2	30773.2	50523	17294124	30773.2



Empty Support Force + the Sum of the Y forces 3449741.50 N
 Operating Support Force + the Sum of the Y forces 3535053.50 N
 Hydro Support Force + the Sum of the Y forces 3493020.00 N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	363528	363528	359241
20	30	347718	347718	343432
30	40	343599	343599	339312
40	50	320111	320111	294076
50	60	198729	198729	191920
60	70	68927.1	68927.1	78544.1
70	80	30773.2	30773.2	50523

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	124238080	124238080	124238080
20	30	124238080	124238080	124238080
30	40	124238080	124238080	124238080
40	50	124238080	124238080	124238080
50	60	124238080	124238080	124238080
60	70	70395464	70395464	70395464
70	80	17294124	17294124	17294124

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N2_400	6.93	10.20	400	600	GR 1.1	2.739	6.931	4.191	6.931
U1_600	6.93	10.20	400	600	GR 1.1	...	...	...	...
K41_50	6.93	10.20	400	600	GR 1.1	...	...	...	...
K1_80	6.93	10.20	400	600	GR 1.1	...	...	...	...
K2_80	6.93	10.20	400	600	GR 1.1	...	...	...	...
K42_80	6.93	10.20	400	600	GR 1.1	...	...	...	...
M1_600	6.93	10.20	400	600	GR 1.1	...	...	...	...
K3_50	6.93	10.20	400	600	GR 1.1	...	...	...	...
K5_80	6.93	10.20	400	600	GR 1.1	...	...	...	...
N1_400	6.93	10.20	400	600	GR 1.1	2.739	6.931	4.191	6.931
Min Rating	6.931	10.201, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N2_400	438.150	0.50	45254832	0	6.432	6.931	Pass
N1_400	438.150	0.50	45254832	0	6.432	6.931	Pass

Evaluating Flange on Nozzle [N2400]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(45254832) + 4 \cdot \text{abs}(0) \cdot 438.150 \leq \pi \cdot 438.150^3 [(6.931 - 6.432) + .500 \cdot 6.931]$$

$$= 0.72408E+09 \leq 0.10478E+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2400 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(6.931, 0.500 \cdot 6.931 + 6.931 - 724077312.0 / (\pi \cdot 438.150^3))$$

$$= \min(6.931, 7.657)$$

$$= 6.931 \text{ MPa}$$

Evaluating Flange on Nozzle [N1400]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(45254832) + 4 \cdot \text{abs}(0) \cdot 438.150 \leq \pi \cdot 438.150^3 [(6.931 - 6.432) + .500 \cdot 6.931]$$

$$= 0.72408E+09 \leq 0.10478E+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1400 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(6.931, 0.500 \cdot 6.931 + 6.931 - 724077312.0 / (\pi \cdot 438.150^3))$$

$$= \min(6.931, 7.657)$$

$$= 6.931 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 8.57863 Hz.

The Natural Frequency for the Vessel (Ope...) is 8.57863 Hz.

The Natural Frequency for the Vessel (Filled) is 9.3735 Hz.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	30334.9	66245552

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	30334.9	66245552

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
70	2182.00	21450.	29700.	21450.	19800.		19800.



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.22 \cdot 0.93) / (1 + 1.7 \cdot 3.4 \cdot 0.22)))$
 $= \min(0.85, 0.89)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	8.579 Hz
Natural Frequency of Vessel (Empty)	8.579 Hz
Natural Frequency of Vessel (Test)	9.374 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.531
Structure Height to Diameter ratio	2.871
Height to top of Structure	13105.000 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1500 + 300 = 1800 \text{ mm}$
 $= 5.91 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
Because z (5.91 ft.) $< 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [0.77 \text{ kPa}]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.53 \cdot 174$$

$$= 1257 \text{ lbs. } [5591 \text{ N}]$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	1800.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-1	3800.0	0.000	0.000	0.000	0.849	1.000	0.766
Bottom head	4300.0	0.000	0.000	0.000	0.849	1.000	0.766
Shell-1	5800.0	0.000	0.000	0.000	0.892	1.000	0.766
Shell-2	8800.0	0.000	0.000	0.000	0.974	1.000	0.766
Shell-2	10650.0	0.000	0.000	0.000	1.014	1.000	0.766
Top head	12020.7	0.000	0.000	0.000	1.040	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.31416E-05 (High); 0.31416E-05 < 0.39270E-05 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0/10700 = 0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[15.6/35.1^2] \cdot 10^4 = 126$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.39270E-05$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 3523640 / (10700 \cdot 4792^2)$$

$$= 0.14341E-04$$

Since Vp is > 0.39270E-05 no further vibration analysis is required !

Platform Load Calculations



ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PL-1	2999999.75	4300.00	0.77	2298.14	1.00
PL-2	2999999.75	7300.00	0.77	2298.14	1.00
PL-3	2999999.75	10300.00	0.77	2298.14	1.00
PL-4	7032000.50	13199.97	0.77	5386.84	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Tailing lug	0.00	300.00	0.77	0.00
Closure plate	0.00	4099.97	0.77	0.00
Hold grid	0.00	4500.00	0.77	0.00
SF wt2	0.00	4300.00	0.77	0.00
ball	0.00	4300.00	0.77	0.00
Clip wt	0.00	8800.00	0.77	0.00
Trunnion 1	0.00	10000.00	0.77	0.00
Trunnion-2	0.00	9999.99	0.77	0.00
misc margin	0.00	7300.00	0.77	0.00
Hold down pla	0.00	10700.00	0.77	0.00
Hold down pla	0.00	10600.00	0.77	0.00
Balls weight	0.00	10500.00	0.77	0.00
Hold down pla	0.00	10300.00	0.77	0.00
SF wt-1	0.00	10300.00	0.77	0.00
N1 wt	0.00	10999.97	0.77	0.00
N1Vane distr	0.00	10999.97	0.77	0.00
M1Davit	0.00	10999.97	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 8.57863 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1800	5388	16163999	0.76606	6149.76
20	30	3800	5388	5388000	0.76606	2049.92
30	40	4299.98	5772	175.931	0.76606	0.066935
40	50	5800	5870.4	17611202	0.76606	8574.53
50	60	8800	5870.4	17611202	0.76606	8797.41
60	70	10650	5870.4	4109280	0.76606	4008.29
70	80	12020.7	5772	10902805	0.76606	10654.9

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(42.30.75)$
 = 0.332 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/8.58)
 = 0.117

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.33$, 0.12) per 12.8.2
 = 0.117

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.16 / (2/2)$
 = 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.1 / (0.12 (2/2))$
 = 0.892

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.16 \cdot 2), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.16 \cdot 3564753$
 = 558953.375 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 3564753$

= 111790.7 N

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 8.57863 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1500	155031	5045.53	5073.38	141491
20	30	3500	40393.4	3067.43	3084.36	136629
30	40	3999.98	230319	19988.7	20099.1	135362
40	50	5500	1190277	142039	139646	128139
50	60	8500	1272833	234740	236036	90812.2
60	70	10350	374137	84017.1	71195.6	50896.2
70	80	10700	301762	70055.8	70442.5	39163.2



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	30334.9	30334.9	390959456	390959456
20	30	3500	30334.9	30334.9	299917920	299917920
30	40	3999.98	30334.9	30334.9	269571680	269571680
40	50	5500	30334.9	30334.9	269570752	269570752
50	60	8500	30334.9	30334.9	178529232	178529232
60	70	10350	30334.9	30334.9	87487672	87487672
70	80	10700	30334.9	30334.9	66245552	66245552

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	40234.9	558953	304829888	4.567E+09
20	30	3500	34085.1	553908	193304672	2.897E+09
30	40	3999.98	32035.2	550840	160232080	2.345E+09
40	50	5500	32035.1	530852	160231104	2.345E+09
50	60	8500	23460.6	388813	76953704	964670272
60	70	10350	14663.2	154073	19744790	150011600
70	80	10700	10654.9	70055.8	10880172	71536816

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	30334.9	0.0016249	0.0062371	0.39831E-05
20	30	30334.9	0.0083703	0.010777	0.50775E-05
30	40	30334.9	0.010777	0.010777	0.50776E-05
40	50	30334.9	0.01868	0.027087	0.57503E-05
50	60	30334.9	0.035897	0.045006	0.61497E-05
60	70	30334.9	0.047163	0.04933	0.62036E-05
70	80	30334.9	0.04933	0.049331	0.62036E-05

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 53.500 Actual deflection : 0.049 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	40234.9	0.001453	0.0054254	0.33643E-05
20	30	34085.1	0.0072138	0.0092022	0.41643E-05
30	40	32035.2	0.0092022	0.0092023	0.41643E-05
40	50	32035.1	0.015641	0.022387	0.45765E-05
50	60	23460.6	0.029339	0.036412	0.47398E-05
60	70	14663.2	0.038073	0.039735	0.47523E-05
70	80	10654.9	0.039735	0.039735	0.47523E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	829.749	5185.93
20	30	829.749	5185.93
30	40	888.885	5555.53
40	50	904.038	5650.24
50	60	904.038	5650.24
60	70	904.038	5650.24
70	80	888.885	5555.53

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 53.500 Actual deflection : 0.040 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 53.500 Actual deflection : 0.089 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	531482	509105	586575808	562076736
20	30	531482	509105	586575808	562076736
30	40	1017564	976406	1110764416	1067254720
40	50	2039258	1998703	2.197E+09	2.156E+09
50	60	2039258	1998703	2.197E+09	2.156E+09
60	70	2039258	1998703	2.197E+09	2.156E+09
70	80	1017564	976406	1110764416	1067254720

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	121.233	280.795	-73.0489	-124.874
20	30	60.6165	140.397	-73.0489	-124.874
30	40	121.233	280.795	-82.464	-139.277
40	50	121.233	280.795	-92.1328	-147.277
50	60	121.233	280.795	-92.1328	-147.277
60	70	121.233	280.795	-92.1328	-147.277
70	80	121.233	280.795	-82.464	-139.277

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	98.7752	-1.69215	170.44
40	50	46.9796	-0.85698	81.0649
50	60	46.9796	-0.85698	81.0649
60	70	46.9796	-0.85698	81.0649
70	80	98.7752	-1.69215	170.44

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-7.00259	-7.00259	-6.92002	0.22096	0.22096
20	30	-6.69805	-6.69805	-6.61548	0.22096	0.22096
30	40	-3.45104	-3.45104	-3.45104	0.11637	0.11637
40	50	-1.57066	-1.57066	-1.57066	0.057604	0.057604
50	60	-0.97508	-0.97508	-0.97508	0.057604	0.057604
60	70	-0.3382	-0.3382	-0.3382	0.03264	0.03264
70	80	-0.30908	-0.30908	-0.30908	0.016199	0.016199

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.22096	0.54215	8.12314	0.27108	...
20	30	0.22096	0.3438	5.15303	0.1719	...
30	40	0.11637	0.15009	2.19628	0.075043	...
40	50	0.057604	0.074293	1.08716	0.037147	...
50	60	0.057604	0.03568	0.44728	0.01784	...
60	70	0.03264	0.0091549	0.069555	0.0045775	...
70	80	0.016199	0.010191	0.067007	0.0050956	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	7.89224
20	30	...	...	...	4.9936
30	40	...	...	...	2.12489
40	50	...	...	...	1.05182
50	60	...	...	...	0.42911
60	70	...	...	...	0.067782
70	80	...	...	...	0.067377

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	0.058343	0.27794
20	30	0.058343	0.26839
30	40	0.03042	0.13864
40	50	0.014861	0.064117
50	60	0.014861	0.045439
60	70	0.014861	0.025467
70	80	0.03042	0.040113

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	0.69534	0.69534	0.6663	0.6663
20	30	0.53342	0.53342	0.51114	0.51114
30	40	0.2525	0.2525	0.24261	0.24261
40	50	0.12499	0.12499	0.12266	0.12266
50	60	0.082777	0.082777	0.081237	0.081237
60	70	0.040565	0.040565	0.03981	0.03981
70	80	0.062051	0.062051	0.059621	0.059621

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		121.23	-8.40	73.05		0.1150
20		60.62	-7.74	73.05		0.1059
30		121.23	-3.94	82.46		0.0478
40		121.23	-1.81	92.13		0.0197
50		121.23	-1.14	92.13		0.0123
60		121.23	-0.41	92.13		0.0044
70		121.23	-0.37	82.46		0.0045

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.08	121.23	-15.53	73.05	0.0172	0.2126
20		60.62	-12.18	73.05		0.1667
30		121.23	-5.81	82.46		0.0704
40		121.23	-2.74	92.13		0.0298
50		121.23	-1.50	92.13		0.0163
60		121.23	-0.45	92.13		0.0049
70		121.23	-0.41	82.46		0.0050

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
-----------	----------------	-------------------	--------------	-------------------	-------------	-------------



10	121.23	-8.40	73.05	0.1150
20	60.62	-7.74	73.05	0.1059
30	121.23	-3.94	82.46	0.0478
40	121.23	-1.81	92.13	0.0197
50	121.23	-1.14	92.13	0.0123
60	121.23	-0.41	92.13	0.0044
70	121.23	-0.37	82.46	0.0045

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.31	121.23	-15.76	73.05	0.0191	0.2158
20		60.62	-12.34	73.05		0.1689
30		121.23	-5.88	82.46		0.0713
40		121.23	-2.78	92.13		0.0301
50		121.23	-1.52	92.13		0.0165
60		121.23	-0.46	92.13		0.0049
70		121.23	-0.41	82.46		0.0050

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		280.79	-7.41	124.87		0.0594
20		140.40	-7.01	124.87		0.0561
30		280.79	-3.64	139.28		0.0262
40		280.79	-1.67	147.28		0.0113
50		280.79	-1.05	147.28		0.0071
60		280.79	-0.38	147.28		0.0025
70		280.79	-0.33	139.28		0.0024

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		280.79	-7.14	124.87		0.0572
20		140.40	-6.84	124.87		0.0547
30		280.79	-3.57	139.28		0.0256
40		280.79	-1.63	147.28		0.0111
50		280.79	-1.03	147.28		0.0070
60		280.79	-0.37	147.28		0.0025
70		280.79	-0.33	139.28		0.0023

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		121.23	-8.40	73.05		0.1150
20		60.62	-7.74	73.05		0.1059
30	95.87	121.23		82.46	0.7908	
40	45.68	121.23		92.13	0.3768	
50	46.20	121.23		92.13	0.3810	
60	46.74	121.23		92.13	0.3855	
70	98.58	121.23		82.46	0.8132	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.31	121.23	-15.76	73.05	0.0191	0.2158
20		60.62	-12.34	73.05		0.1689
30	98.03	121.23		82.46	0.8086	
40	46.74	121.23		92.13	0.3856	
50	46.64	121.23		92.13	0.3847	
60	46.81	121.23		92.13	0.3861	
70	98.65	121.23		82.46	0.8137	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
-----------	----------------	-------------------	--------------	-------------------	-------------	-------------



Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10		121.23	-8.40	121.23		0.0693
20		60.62	-7.74	73.05		0.1059
30		121.23	-5.63	82.46		0.0683
40		121.23	-2.67	92.13		0.0290
50		121.23	-1.99	92.13		0.0216
60		121.23	-1.26	92.13		0.0137
70		121.23	-2.06	82.46		0.0250

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.31	121.23	-15.76	121.23	0.0191	0.1300
20		60.62	-12.34	73.05		0.1689
30		121.23	-7.57	82.46		0.0918
40		121.23	-3.63	92.13		0.0394
50		121.23	-2.37	92.13		0.0258
60		121.23	-1.31	92.13		0.0142
70		121.23	-2.11	82.46		0.0255

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		280.79	-7.41	124.87		0.0594
20		140.40	-7.01	124.87		0.0561
30	167.18	280.79		139.28	0.5954	
40	79.59	280.79		147.28	0.2834	
50	80.17	280.79		147.28	0.2855	
60	80.76	280.79		147.28	0.2876	
70	170.15	280.79		139.28	0.6060	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		280.79	-7.14	124.87		0.0572
20		140.40	-6.84	124.87		0.0547
30	167.11	280.79		139.28	0.5951	
40	79.55	280.79		147.28	0.2833	
50	80.15	280.79		147.28	0.2854	
60	80.76	280.79		147.28	0.2876	
70	170.15	280.79		139.28	0.6059	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		121.23	-7.22	73.05		0.0989
20		60.62	-6.92	73.05		0.0947
30	95.44	121.23		82.46	0.7872	
40	45.47	121.23		92.13	0.3750	
50	46.06	121.23		92.13	0.3799	
60	46.67	121.23		92.13	0.3850	
70	98.48	121.23		82.46	0.8123	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		121.23	-6.71	73.05		0.0918
20		60.62	-6.42	73.05		0.0878
30	95.46	121.23		82.46	0.7874	
40	45.44	121.23		92.13	0.3748	
50	46.02	121.23		92.13	0.3796	
60	46.65	121.23		92.13	0.3848	
70	98.48	121.23		82.46	0.8123	



Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		121.23	-7.22	73.05		0.0989
20		60.62	-6.92	73.05		0.0947
30	95.44	121.23		82.46	0.7872	
40	45.47	121.23		92.13	0.3750	
50	46.06	121.23		92.13	0.3799	
60	46.67	121.23		92.13	0.3850	
70	98.48	121.23		82.46	0.8123	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		121.23	-7.22	73.05		0.0989
20		60.62	-6.92	73.05		0.0947
30	95.44	121.23		82.46	0.7872	
40	45.47	121.23		92.13	0.3750	
50	46.06	121.23		92.13	0.3799	
60	46.67	121.23		92.13	0.3850	
70	98.48	121.23		82.46	0.8123	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		121.23	-7.22	73.05		0.0989
20		60.62	-6.92	73.05		0.0947
30		121.23	-3.57	82.46		0.0433
40		121.23	-1.63	92.13		0.0177
50		121.23	-1.03	92.13		0.0112
60		121.23	-0.37	92.13		0.0040
70		121.23	-0.33	82.46		0.0039

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		101.03	-7.64	60.87		0.1255
20		50.51	-7.18	60.87		0.1180
30	95.83	101.03		68.72	0.9486	
40	45.66	101.03		76.78	0.4519	
50	46.19	101.03		76.78	0.4572	
60	46.74	101.03		76.78	0.4626	
70	98.58	101.03		68.72	0.9758	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		101.03	-7.64	101.03		0.0756
20		50.51	-7.18	60.87		0.1180
30		101.03	-5.37	68.72		0.0782
40		101.03	-2.55	76.78		0.0332
50		101.03	-1.93	76.78		0.0251
60		101.03	-1.24	76.78		0.0162
70		101.03	-2.04	68.72		0.0297

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		121.23	-7.22	73.05		0.0989
20		60.62	-6.92	73.05		0.0947
30	95.44	121.23		82.46	0.7872	
40	45.47	121.23		92.13	0.3750	
50	46.06	121.23		92.13	0.3799	
60	46.67	121.23		92.13	0.3850	



70 98.48 121.23 82.46 0.8123

Absolute Maximum of the all of the Stress Ratio's 0.9758

Governing Element: Top head

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-2	...	-6.7	0.2	0.7	...	+ Tens
Skirt-2	...	-6.7	-0.2	-0.7	-7.6	- Comp
Skirt-1	...	-6.4	0.2	0.5	...	+ Tens
Skirt-1	...	-6.4	-0.2	-0.5	-7.2	- Comp
Bottom head	98.8	-3.3	0.1	0.3	95.8	+ Tens
Bottom head	98.8	-3.3	-0.1	-0.3	...	- Comp
Shell-1	47.0	-1.5	0.1	0.1	45.7	+ Tens
Shell-1	47.0	-1.5	-0.1	-0.1	...	- Comp
Shell-2	47.0	-0.9	0.1	0.1	46.2	+ Tens
Shell-2	47.0	-0.9	-0.1	-0.1	...	- Comp
Shell-2	47.0	-0.3	...	...	46.7	+ Tens
Shell-2	47.0	-0.3	...	...	...	- Comp
Top head	98.8	-0.3	...	0.1	98.6	+ Tens
Top head	98.8	-0.3	...	-0.1	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

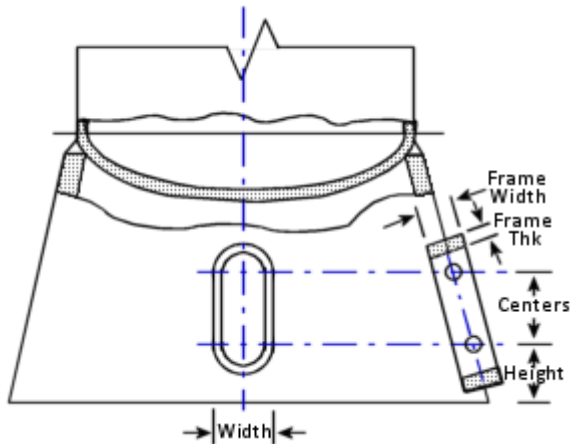
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-2

Total vessel (empty) weight	Fg	3384038.250 N
Weight of vessel contents (liquid)	Ff	0.000 N
Vessel Gross Weight	Fg + Ff	3384038.250 N
Skirt Mean Diameter	Dz	4452.000 mm
Skirt Analysis Thickness (Corroded)	ez	36.400 mm
Skirt Height	H	3000.000 mm
Additional Axial Force at section 1-1	F1	29700.000 N
Moment at section 1-1	M1	3197242112.000 N-mm
Applied Moment - Bottom of Skirt	Mb	4958252544.000 N-mm
Weight at Bottom of Skirt	W	3564753.750 N



Access Opening: 1 Cuts through the top of the skirt.
 Access Opening: 3 Cuts through the top of the skirt.
 Access Opening: 4 Cuts through the top of the skirt.
 Access Opening: 6 Cuts through the top of the skirt.

Note:

If an opening cuts the skirt top or bottom, this is not a problem.
 This becomes a 'Sentry Box' or 'Mouse Hole' type Opening.

Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	50.8000	0.0	3000.00	0.00	50.000	16.000
2	600.0000	0.0	1800.00	0.00	50.000	16.000
3	50.8000	90.0	3000.00	0.00	50.000	16.000
4	50.8000	180.0	3000.00	0.00	50.000	16.000
5	700.0000	180.0	2000.00	0.00	50.000	16.000
6	50.8000	270.0	3000.00	0.00	50.000	16.000

Interference check:



Access	Access	Pitch mm	Ligament mm
1	2	1200.0000	874.6000
2	3	3696.7212	3371.3213
3	4	3496.5337	3445.7339
4	5	1000.0000	624.6000
5	6	3636.7222	3261.3223
1	6	3496.5337	3445.7339

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 1000.0000 mm
 Minimum Ligament between Openings is: 624.6000 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(4452 \cdot 36.4)} \text{ mm}$$

$$= 805.116 \text{ mm}$$

Carefully evaluate this acceptance criteria.



Ligament is smaller than IMin - Evaluate carefully

Section properties at least metal area [A]:

Number of Openings At Critical Height = 5

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
2	-7.3786	7.3786
5	171.3859	188.6141

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
0.000	-7.379
7.379	0.000
0.000	0.000
0.000	171.386
-171.386	0.000

Height of Weakest Section	H	1893.515	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	38.991	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	1260141281280.000	mm ⁴
Principle Moment of Inertia	Iyy	1077281554432.000	mm ⁴
Distance to Extreme Fiber	ybar	2244.200	mm
Distance to Extreme Fiber	xbar	2283.191	mm
Minimum Section Modulus	W4	471832032.000	mm ³
Metal Area at the Cutting Plane	A4	471870.812	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(700/4452)$$

$$= 0.16 \text{ rad (9.05 deg)}$$

Note: The largest opening width is considered.

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 4452/2$$

$$= 2226.0 \text{ mm}$$



Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
1600.0	1266.7	...	...
1600.0	21319.0	...	...
1600.0	1266.7	...	...
1600.0	1266.7	...	...
1600.0	24996.6	...	...
1600.0	1266.7	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.16 \cdot \sqrt{(2226/36.4)}$$

$$= 1.235$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= ex(new) - ci - cext$$

$$= 38 - 0.8 - 0.8$$

$$= 36.4 \text{ mm}$$

Note: At least one of the openings cuts the Top Of the Skirt (Ok)

Find the moment acting at the Critical Height [M4]:

$$= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top}$$

$$= (4.96E+09 - 3.20E+09) \cdot (3000 - 1894) / 3000 + 3.20E+09$$

$$= 3.85E+09 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{bar} = 1.26E+12 / 2244 = 561510784 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{bar} = 1.08E+12 / 2283 = 471832032 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(561510784, 471832032) = 471832032 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / Area(A_4))}$$

$$= \sqrt{(\min(1.26E+12, 1.08E+12) / 471871)}$$

$$= 1511 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 1511 / 3000$$

$$= 0.504$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(39^2 + -0.66238E-06^2)}$$

$$= 38.991 \text{ mm}$$

$$\text{Where } e_x = 39 : e_y = -0.66238E-06$$

Moment Increment at Centroid [ΔM_4]:

$$= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e$$

$$= \text{abs}(29700 - 3384038 - 0) \cdot 39$$

$$= 130841496.000 \text{ N-mm}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:



Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned} &= \min(1; A4/(\pi \cdot D3 \cdot ea3)) \\ &= \min(1; 471871/(\pi \cdot 4452 \cdot 36.4)) \\ &= 0.927 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned} &= \min(1; 4 \cdot W4/(\pi \cdot D^2/3/(\pi \cdot D3 \cdot ea3))) \\ &= \min(1; 4.0 \cdot 471832032/(\pi \cdot 4452^2 \cdot 36.4)) \\ &= 0.833 \end{aligned}$$

Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 4452^2 \cdot 36.4 \cdot 73 \\ &= 41405157376.0 \text{ N-mm} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call} \\ &= \pi \cdot 4452 \cdot 36.4 \cdot 73 \\ &= 37186328.0 \text{ N} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F4/(\psi1 \cdot F_{c,max}) + (abs(M4) + F4 \cdot abs(yq))/(\psi2 \cdot Mmax) \\ &= 3354338/(0.93 \cdot 37186328) + (abs(3.85E+09) + 3354338 \cdot abs(39))/ \\ &\quad (0.83 \cdot 41.41E+09) \\ &= 0.213 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Skirt Support Analysis Per EN 13445-3 Section 16.12:

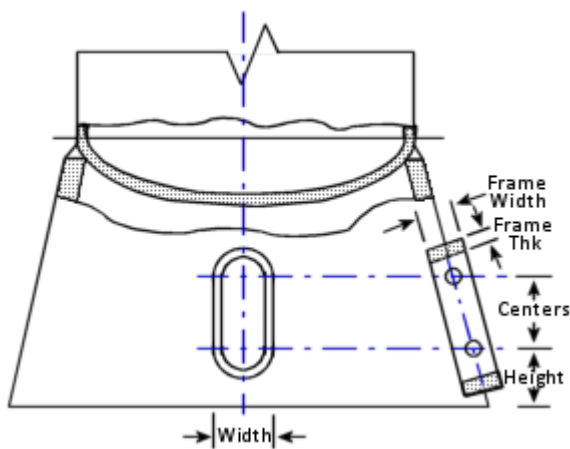
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

Total vessel (empty) weight	Fg	3384038.250 N
Head Weight below section 2-2	DeltaFg	230319.438 N
Weight of vessel contents (liquid)	Ff	0.000 N
Vessel Gross Weight	Fg + Ff	3384038.250 N
Skirt Mean Diameter	Dz	4452.000 mm
Skirt Analysis Thickness (Corroded)	ez	36.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	29700.000 N
Moment at section 1-1	M1	2614314496.000 N-mm
Applied Moment - Bottom of Skirt	Mb	3197242112.000 N-mm
Weight at Bottom of Skirt	W	3409722.500 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Spherical
Head Mean Diameter at Skirt Top	Db	4440.000 mm
Head Corroded Thickness	eb	70.000 mm



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	1261322895360.000 mm ⁴
Moment of Inertia	Iyy =	1261322895360.000 mm ⁴
Section Modulus	W =	566632576.000 mm ³
Section Metal Area	A =	509103.906 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\frac{\min(I_{xx}, I_{yy})}{\text{Area}}}$$

$$= \sqrt{\frac{\min(1.26\text{E}+12, 1.26\text{E}+12)}{509104}}$$

$$= 1574 \text{ mm}$$

Skirt Slenderness Ratio [rg/L]:

$$= \text{rg} / \text{L}$$

$$= 1574/1000$$

$$= 1.574$$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 2.61E+09 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -2.61E+09 / 2.27E+09 - 3409722 / 509104$$

$$= -7.85 \text{ MPa}$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +2.61E+09 / 2.27E+09 - 3409722 / 509104$$

$$= -5.54 \text{ MPa}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= 29700 - 3384038 - 0 + 4 \cdot 2.61E+09 / 4452$$

$$= -1006400 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= 29700 - 3384038 - 0 - 4 \cdot 2.61E+09 / 4452$$

$$= -5702276 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-1006400 + 230319 + 0) / (\pi \cdot 4440 \cdot 70) + 6.41 \cdot 4440 / (4 \cdot 70)$$

$$= 100.87 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-5702276 + 230319 + 0) / (\pi \cdot 4440 \cdot 70) + 6.41 \cdot 4440 / (4 \cdot 70)$$

$$= 96.06 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (0 + 230319) / (\pi \cdot 4440 \cdot 70) + 6.41 \cdot 4440 / (4 \cdot 70)$$

$$= 101.90 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= -1006400 / (\pi \cdot 4452 \cdot 36.4)$$

$$= -1.98 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -5702276 / (\pi \cdot 4452 \cdot 36.4)$$

$$= -11.20 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(4440 + 70 - 4452 + 36.4) / (2185 + 70)$$

$$= 0.98$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.98) = 11.7 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(70^2 + 36.4^2 + 2 \cdot 70 \cdot 36.4 \cdot 0.98)}$$

$$= 52.9 \text{ mm}$$



Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 52.94 \cdot -1006400.3 = -53.3093 \times 10^6 \text{ N-mm} \\ M_q &= a \cdot F_{zq} = 52.94 \cdot -5702276.0 = -57.02276 \times 10^6 \text{ N-mm} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{70}{36.4} \right)^2 \\ &= 0.419 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ &= 0.42 \cdot 6 \cdot -53309276 / (\pi \cdot 4440 \cdot 70^2) \\ &= -1.96 \text{ MPa} \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ &= 0.42 \cdot 6 \cdot -302050976 / (\pi \cdot 4440 \cdot 70^2) \\ &= -11.11 \text{ MPa} \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ &= 0.42 \cdot 6 \cdot -53309276 / (\pi \cdot 4452 \cdot 36.4^2) \\ &= -7.23 \text{ MPa} \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ &= 0.42 \cdot 6 \cdot -302050976 / (\pi \cdot 4452 \cdot 36.4^2) \\ &= -40.98 \text{ MPa} \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} eb/D_b &= 2.76/175 = 0.016 \\ y &= 125 \cdot eb/D_b = 125 \cdot 70/4440 = 1.97 \\ \alpha &= 0 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ MPa (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 101 - -1.96 + 0 \\ &= 102.83 \text{ MPa} \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 101 + -1.96 - 0 \\ &= 98.90 \text{ MPa} \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 96.1 - -11.1 + 0 \\ &= 107.17 \text{ MPa} \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 96.1 + -11.1 - 0 \\ &= 84.94 \text{ MPa} \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 102 + -1.96 + 0 \\ &= 99.93 \text{ MPa} \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 102 - -1.96 - 0 \\
 &= 103.86 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qitot}]: \\
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 102 + -11.1 + 0 \\
 &= 90.78 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qotot}]: \\
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 102 - -11.1 - 0 \\
 &= 113.01 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3pitot}]: \\
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= -1.98 - -7.23 \\
 &= 5.26 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3potot}]: \\
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= -1.98 + -7.23 \\
 &= -9.21 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qitot}]: \\
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -11.2 - -41 \\
 &= 29.78 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}]: \\
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -11.2 + -41 \\
 &= -52.18 \text{ MPa}
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 101 (3 - (1/1.5)) \cdot (101/101)^2 \\
 &= 235.95 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 101 (3 - (1/1.5)) \cdot (96.1/101)^2 \\
 &= 242.20 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 101 (3 - (1/1.5)) \cdot (102/101)^2 \\
 &= 234.57 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 101 (3 - (1/1.5)) \cdot (102/101)^2 \\
 &= 234.57 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 101 (3 - (1/1.5)) \cdot (-1.98/101)^2 \\
 &= 303.06 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 101 (3 - (1/1.5)) \cdot (-11.2/101)^2 \\
 &= 302.25 \text{ MPa}
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	102.83	235.95	Pass	46
Stot1po	98.90	235.95	Pass	47
Stot1qi	107.17	242.20	Pass	48
Stot1qo	84.94	242.20	Pass	49
Stot2pi	99.93	234.57	Pass	50
Stot2po	103.86	234.57	Pass	51
Stot2qi	90.78	234.57	Pass	52
Stot2qo	113.01	234.57	Pass	53
Stot3pi	5.26	303.06	Pass	54
Stot3po	-9.21	303.06	Pass	55
Stot3qi	29.78	302.25	Pass	56
Stot3qo	-52.18	302.25	Pass	57

The stresses are satisfactory in the region without openings.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Shop/Field Installation Options :**

Platform(s) installed in the Shop.

Packing is installed in the Shop.

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	8200.876 mm
Center of Gravity of the Packing	7350.000 mm
Center of Gravity of the Insulation	7349.987 mm
Center of Gravity of the Nozzles	8374.735 mm
Center of Gravity of the Added Weights (Operating)	7650.712 mm
Center of Gravity of the Added Weights (Empty)	7458.996 mm
Center of Gravity of Bare Shell New and Cold	6738.371 mm
Center of Gravity of Bare Shell Corroded	6745.602 mm
Vessel CG in the Operating Condition	7165.822 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	7130.323 mm
Vessel CG in the Test Condition	7192.713 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	4490.0000	mm
Skirt Thickness	STHK	38.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	38.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-36	
Basing Ope. All. Stress	BASOPE, f4	114.00	MPa
Basing Yield Stress		248.00	MPa
Inside Diameter of Basing	DI, D4	4274.0000	mm
Outside Diameter of Basing	DOU, D5	4770.0000	mm

Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	138.00	MPa
Number of Bolts	RN, nb	24	
Diameter of Bolt Circle	DC, DBC	4630.0000	mm
Bolt Allowable Shear Stress		82.800	MPa

Ultimate Comp. Strength of Concrete	FPC	41.0	MPa
Allowable Comp. Strength of Concrete	FC	22.5	MPa
Modular ratio Steel/Concrete		9.833	

Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	140.0000	mm
Width of Gussets at Base Plate	BWDT	140.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	248.0	MPa
Height of Chair	h1	346.0000	mm
Distance between Gussets	RG, b6	96.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	44.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm

External Corrosion Allowance	CA	1.6000	mm
------------------------------	----	--------	----

Dead Weight of Vessel	DW	3449741.8	N
Operating Weight of Vessel	ROW	3535053.5	N
Earthquake Moment on Basing	EQMOM	4635941888.0	N-mm
Wind Moment on Basing	WIMOM	373478624.0	N-mm

User Moment at the Base [Earthquake Case]		390959456.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		390959456.0	N-mm (UWIMOM)
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
-----------------------------------	------	--------	--

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 5.03E+09/4630) - 3423262)/24$$

$$= 38244.7 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 764438080/4630) - 3449741)/24$$

$$= -116232.8 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/4630) - 3535053)/24$$

$$= -147293.9 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	1155.1653	mm ²
Area Available in all the Bolts	Abss * RN :	27723.9668	mm ²



*There seems to be no tensile load on bolts per the Brownell and Young Method.
 It is possible that this method is encountering a problem. It is strongly
 recommended that you try the simplified analysis method for the basing.*

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3)/(2kd)]$$

$$= 1.78[(2 \cdot 1 \cdot 4630 + 248)/(2 \cdot 1 \cdot 4630)]$$

$$= 1.83 \text{ MPa, Should be } \leq 22.5 \text{ MPa}$$

Values for table 10.3, l = 140, b = 303, l/b = 0.46199614

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(4089, 12299) = 12299 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 12299/164)} + 1.6$$

$$= 22.8 \text{ mm}$$

Nomenclature:

a = (Dc-Ds)/2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 140, b = 96, b/l = 0.6857143

As b/l (0.69) is less than 1, inverting b/l = 1.46 0.

Moment Term, based on geometry [Mo]:

$$= P/(4 \cdot \pi) [1.3(\ln((2\ln(\pi \cdot a/l)/(\pi \cdot g))) + 1) - [(0.7 - g2)P/(4\pi)]]$$

$$= 159399/(4 \cdot \pi) [1.3(\ln((2 \cdot 140 \cdot \sin(\pi \cdot 70/140)) /$$

$$(\pi \cdot 40))) + 1] - [(0.7 - 0.076) \cdot 159399/(4 \cdot \pi)]$$

$$= 17974.5332 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot Abs(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot Abs(17975) / 165)} + 1.6$$

$$= 27.1415 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot lavg) + Ca$$

$$= 159399/(124 \cdot 140) + 1.6$$

$$= 11.1 \text{ (not less than } 9.52 + 1.6) \text{ mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 4630/24 = 606 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot Sktope))^{2/3} \cdot r^{1/3} + Ca$$

$$= 1.76(159399 \cdot 70 / (606 \cdot 302 \cdot 152))^{2/3} \cdot 22451^{1/3} + Ca$$

$$= 12.6 + 1.6 = 14.2 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 589288 / (1155 \cdot 24)$$

$$= 21.3 \text{ MPa. Must be less than 82.8 MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	22.8342	mm
Actual Basing Thickness as entered by user	38.0000	mm
Required Thickness of Chair Cap	27.1415	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	11.1250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	14.1592	mm
Given Thickness of Skirt	38.0000	mm
Required Gusset Height to meet local stress	23.2062	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$$

$$= 15.8 \cdot (38 - 1.6)$$

$$= 574 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 574 / [(0.4 \cdot 181) \cdot 2 \cdot 0.707]$$

$$= 5.6 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 159399 / (180 + 2 \cdot (264 + 44))$$

$$= 200 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 159399 \cdot 70 / (180 \cdot (308) + 0.6667 \cdot (308)^2)$$

$$= 94 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{ Wv^2 + Wh^2 }$$

$$= \sqrt{ 200^2 + 94^2 }$$

$$= 221 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 221 / [(0.4 \cdot 181) \cdot 2 \cdot 0.707]$$

$$= 2.16 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 159399 / (2 \cdot 140)$$

$$= 569 \text{ N/mm}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 159399 \cdot 70 / (2 \cdot 308 \cdot 140) \\ &= 129 \text{ N/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{ Wv^2 + Wh^2 } \\ &= \sqrt{ 569^2 + 129^2 } \\ &= 584 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 584 / [(0.4 \cdot 181) \cdot 2 \cdot 0.707] \\ &= 5.7 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	5.7030	mm

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N2400

From: 30

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	4364.00	mm
Head Finished (Minimum) Thickness	t	73.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		400.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

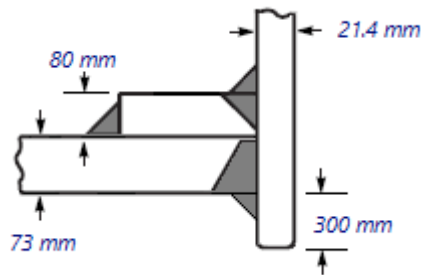
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	73.0000	mm
Inside Projection	h	300.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	806.4000	mm
Thickness of Pad	te	80.0000	mm
Weld leg size between Pad and Shell	Wp	80.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	80.0000	mm
Reinforcing Pad Width		200.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2500.0	mm
Centerline Distance to Tangent		2800.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2400

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	406.400 mm.
Actual Thickness Used in Calculation	21.412 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
 $= 69.7705 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.41 \cdot 203 / (88.9 \cdot 1 + 0.4 \cdot 6.41)$
 $= 14.2429 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0528 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	739.1512 mm
Parallel to Vessel Wall, opening length	d	369.5756 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		126.0305 mm
Normal to Vessel Wall, Inward		38.5305 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.88, 1)$
 $= 0.880$

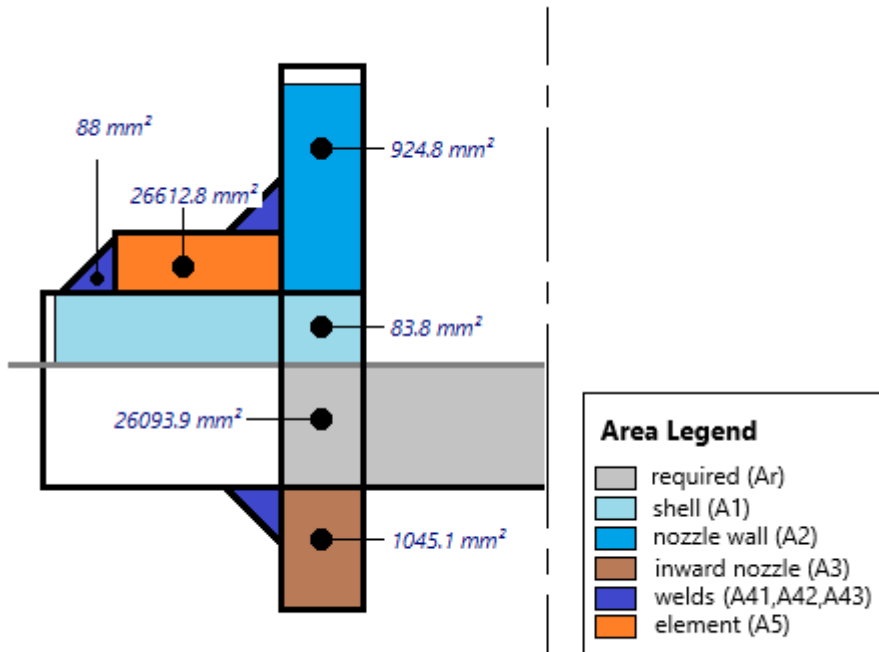


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26093.904	1344.273	NA
Area in Shell	A1	83.796	22935.869	NA
Area in Nozzle Wall	A2	924.772	3850.369	NA
Area in Inward Nozzle	A3	1045.108	1045.108	NA
Area in Welds	A41+A42+A43	87.996	87.996	NA
Area in Element	A5	26612.834	26612.834	NA
TOTAL AREA AVAILABLE	Atot	28754.504	54532.172	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 705.8846 80.0000 mm
 Based on given Pad Diameter: 806.4001 72.0021 mm
 Based on the Estimated Diameter Limit: 737.5637 72.3472 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (370 \cdot 69.8 \cdot 1 + 2 \cdot 18.4 \cdot 69.8 \cdot 1 \cdot (1 - 0.88))$
 $= 26093.904 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 370(1 \cdot 70 - 1 \cdot 69.8) - 2 \cdot 18.4$
 $(1 \cdot 70 - 1 \cdot 69.8) \cdot (1 - 0.88)$
 $= 83.796 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 126) \cdot (18.4 - 14.2) \cdot 0.88$

= 924.772 mm²

Area Available in Inward Nozzle [A3]:

$$\begin{aligned} &= 2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2 \\ &= 2 \cdot 15.4 \cdot (38.5) \cdot 0.88 \\ &= 1045.108 \text{ mm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr3 + (W_i\text{-can}/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4 \\ &= 10^2 \cdot 0.88 + (0)^2 \cdot 0.88 + 0^2 \cdot 1 \\ &= 87.996 \text{ mm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (739 - 406) \cdot 80 \cdot 1 \\ &= 26612.834 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.2429 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 72.7705 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 72.7705 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.1185 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.3, \max(72.8, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(17.2, 11.3) \\ &= 17.2429 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 21.412 = 18.736 \text{ mm}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 54.0,	Allowable	: 88.9 MPa	Passed
Expansion	: 0.0,	Allowable	: 204.7 MPa	Passed
Occasional	: 30.6,	Allowable	: 118.2 MPa	Passed
Shear	: 12.8,	Allowable	: 62.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 18.7, tr = 10.8, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	10 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-7 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 18.7, tr = 10.8, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 18.7, tr = 10.8, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 73$, $tr = 51$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.73$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 25 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 10 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 18.7$, $tr = 10.8$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Gov. MDMT of the Nozzle : -7 °C
 Gov. MDMT of the Reinforcement Pad : 10 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 10 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.41/10.20 = 0.628$

Weld Size Calculations, Description: N2400

Intermediate Calc. for nozzle/shell Welds Tmin 18.4122 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 * \text{TminPad}$	$56.5600 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (26094 - 83.8 + 2 \cdot 18.4 \cdot 0.88 \cdot (1 \cdot 70 - 69.8)) \cdot 101)$
 $= 2628266.00 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (925 + 26613 + 88 - 0 \cdot 0.88) \cdot 101$
 $= 2790710.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (925 + 1045 + 88 + (2268)) \cdot 101$
 $= 437023.25 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (925 + 1045 + 88 + 26613 + (2268)) \cdot 101$
 $= 3125424.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 406 \cdot 10 \cdot 0.49 \cdot 88.9$
 $= 278057. N$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 806 \cdot 80 \cdot 0.49 \cdot 101$
 $= 5014654. N$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 194) \cdot (21.4 - 3) \cdot 0.7 \cdot 88.9$
 $= 698243. N$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 406 \cdot 80 \cdot 0.74 \cdot 101$
 $= 3816627. N$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 406 \cdot (73 - 3) \cdot 0.74 \cdot 101$
 $= 3340461. N$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (5014654 + 698243) = 5712897 N$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (278057 + 3816627 + 3340460 + 0) = 7435145 N$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (5014654 + 3340460 + 0) = 8355115 N$

Summary of Failure Path Calculations:

Path 1-1 = 5712897 N , must exceed W = 2628266 N or W1 = 2790710 N
 Path 2-2 = 7435145 N , must exceed W = 2628266 N or W2 = 437023 N
 Path 3-3 = 8355115 N , must exceed W = 2628266 N or W3 = 3125424 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.43 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 9.4822 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 573.0000 mm

Input Echo, WRC107/537 Item 1, Description: N2400 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	4364.000	mm
Vessel Thickness	Tv	73.000	mm
Design Temperature	T1	400.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	101.03	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

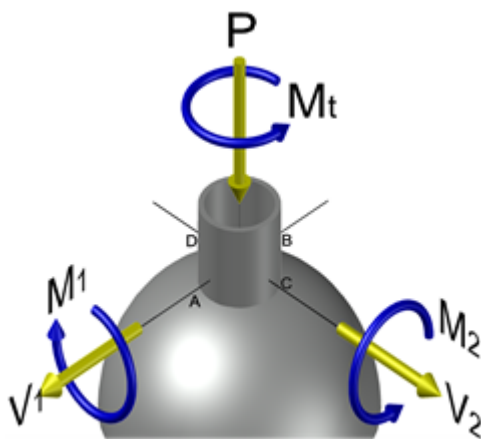
Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	3.0000 mm
Nozzle Diameter	Dn	406.400 mm
Nozzle Thickness	Tn	21.412 mm
Nozzle Material		SA-106 B
Nozzle UNS Number		K03006
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	88.90 MPa
Thickness of Reinforcing Pad	Tpad	80.000 mm
Diameter of Reinforcing Pad	Dpad	806.400 mm
Design Internal Pressure	Dp	6.411 MPa
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	36000.0	N
Longitudinal Shear (SUS)	(V1) V1	26000.0	N
Circumferential Shear (SUS)	(Vc) V2	26000.0	N
Circumferential Moment (SUS)	(Mc) M1	32000000.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	32000000.0	N-mm
Torsional Moment (SUS)	Mt	36799944.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979
Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(2220.0 (73.0 - 3.0))}$$

$$= 1707.286 \text{ mm}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	36000.0	N
Circumferential Shear	(VC) V2	26000.0	N
Longitudinal Shear	(VL) V1	26000.0	N



Circumferential Moment (MC) M1 32000000.0 N-mm
 Longitudinal Moment (ML) M2 32000000.0 N-mm
 Torsional Moment MT 36799944.0 N-mm

Dimensionless Param: U = 0.35 TAU = 10.54 RHO = 4.00 (8.15)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.05943	(A,B,C,D)
M(x) / P	SP 4	0.02342	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.17430	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.07451	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.17430	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.07451	(A,B,C,D)
N(y) * T / P	SP 4	0.33512	(A,B,C,D)
M(y) / P	SP 4	0.11820	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.28954	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	0.43328	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.28954	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	0.43328	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad.	Bend. P	-0.2	0.2	-0.2	0.2	-0.2	0.2	-0.2	0.2
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-1.1	1.1	1.1	-1.1
Rad.	Memb. ML	-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-1.1	1.1	1.1	-1.1	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-1.8	0.8	1.2	-0.5	-1.8	0.8	1.2	-0.5
Tang.	Memb. P	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Tang.	Bend. P	-1.1	1.1	-1.1	1.1	-1.1	1.1	-1.1	1.1
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-6.3	6.3	6.3	-6.3
Tang.	Memb. ML	-0.7	-0.7	0.7	0.7	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-6.3	6.3	6.3	-6.3	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-8.7	6.2	5.4	-5.0	-8.7	6.2	5.4	-5.0
	Shear VC	0.3	0.3	-0.3	-0.3	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
	Shear MT	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tot. Shear		1.2	1.2	0.7	0.7	0.7	0.7	1.2	1.2
Str. Int.		8.9	6.5	5.5	5.1	8.8	6.3	5.7	5.4

Unitless Prm: U = 1.02 TAU = 0.00 (21.40) RHO = 0.00 (3.80)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.06388	(A,B,C,D)
M(x) / P	SR 2	0.03343	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.07408	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.06860	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.07408	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.06860	(A,B,C,D)

N(y) * T / P	SR 2	0.01935	(A,B,C,D)
M(y) / P	SR 2	0.01031	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.02196	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.02069	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.02196	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.02069	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Rad. Bend. P		-1.5	1.5	-1.5	1.5	-1.5	1.5	-1.5	1.5
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Rad. Bend. MC		0.0	0.0	0.0	0.0	-6.8	6.8	6.8	-6.8
Rad. Memb. ML		-1.2	-1.2	1.2	1.2	0.0	0.0	0.0	0.0
Rad. Bend. ML		-6.8	6.8	6.8	-6.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-10.0	6.6	6.1	-4.6	-10.0	6.6	6.1	-4.6
Tang. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Tang. Bend. P		-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Tang. Bend. MC		0.0	0.0	0.0	0.0	-2.1	2.1	2.1	-2.1
Tang. Memb. ML		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Tang. Bend. ML		-2.1	2.1	2.1	-2.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-3.0	2.0	1.8	-1.4	-3.0	2.0	1.8	-1.4
Shear VC		0.3	0.3	-0.3	-0.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tot. Shear		0.8	0.8	0.2	0.2	0.2	0.2	0.8	0.8
Str. Int.		10.1	6.7	6.1	4.6	10.0	6.6	6.2	4.8

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		46.8	46.8	46.8	46.8	46.8	46.8	46.8	46.8
Rad. Pl (SUS)		-0.5	-0.5	0.3	0.3	-0.5	-0.5	0.3	0.3
Rad. Q (SUS)		-1.3	1.3	0.9	-0.9	-1.3	1.3	0.9	-0.9
Long. Pm (SUS)		46.8	46.8	46.8	46.8	46.8	46.8	46.8	46.8
Long. Pl (SUS)		-1.2	-1.2	0.2	0.2	-1.2	-1.2	0.2	0.2
Long. Q (SUS)		-7.5	7.5	5.2	-5.2	-7.5	7.5	5.2	-5.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.3	0.3	-0.3	-0.3	-0.3	-0.3	0.3	0.3
Shear Q (SUS)		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Pm (SUS)		46.8	46.8	46.8	46.8	46.8	46.8	46.8	46.8
Pm+Pl (SUS)		46.4	46.4	47.4	47.4	46.4	46.4	47.4	47.4
Pm+Pl+Q (Total)		45.2	53.3	52.3	46.4	45.1	53.2	52.5	46.6



Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.84	101.03	Passed
Pm+Pl (SUS)	47.37	151.54	Passed
Pm+Pl+Q (TOTAL)	53.33	303.08	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	100.1	100.1	100.1	100.1	100.1	100.1	100.1	100.1
Rad.	Pl (SUS)	-1.7	-1.7	0.8	0.8	-1.7	-1.7	0.8	0.8
Rad.	Q (SUS)	-8.3	8.3	5.3	-5.3	-8.3	8.3	5.3	-5.3
Long.	Pm (SUS)	100.1	100.1	100.1	100.1	100.1	100.1	100.1	100.1
Long.	Pl (SUS)	-0.5	-0.5	0.2	0.2	-0.5	-0.5	0.2	0.2
Long.	Q (SUS)	-2.5	2.5	1.6	-1.6	-2.5	2.5	1.6	-1.6
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.3	0.3	-0.3	-0.3	-0.3	-0.3	0.3	0.3
Shear	Q (SUS)	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Pm (SUS)		100.1	100.1	100.1	100.1	100.1	100.1	100.1	100.1
Pm+Pl (SUS)		99.7	99.7	101.0	101.0	99.7	99.7	101.0	101.0
Pm+Pl+Q (Total)		97.2	106.9	106.2	98.8	97.1	106.7	106.4	98.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	100.12	101.03	Passed
Pm+Pl (SUS)	101.01	151.54	Passed
Pm+Pl+Q (TOTAL)	106.86	303.08	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: U1600

From: 30

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	4364.00	mm
Head Finished (Minimum) Thickness	t	73.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
-------------------------------	----	-----------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	101.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

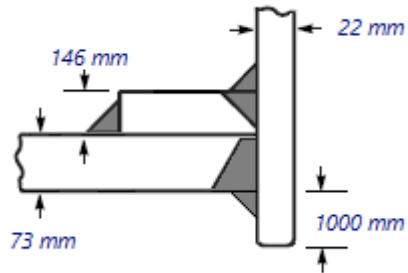
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	74.0000	mm
Inside Projection	h	1000.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	840.0000	mm
Thickness of Pad	te	146.0000	mm
Weld leg size between Pad and Shell	Wp	110.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	146.0000	mm
Reinforcing Pad Width		120.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: U1600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	600.000 mm.
Actual Thickness Used in Calculation	22.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
 $= 69.7705 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.41 \cdot 300 / (101 \cdot 1 + 0.4 \cdot 6.41)$
 $= 18.5711 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 3.0634 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1124.0001 mm
Parallel to Vessel Wall, opening length	d	562.0001 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		175.0000 mm
Normal to Vessel Wall, Inward		40.0000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

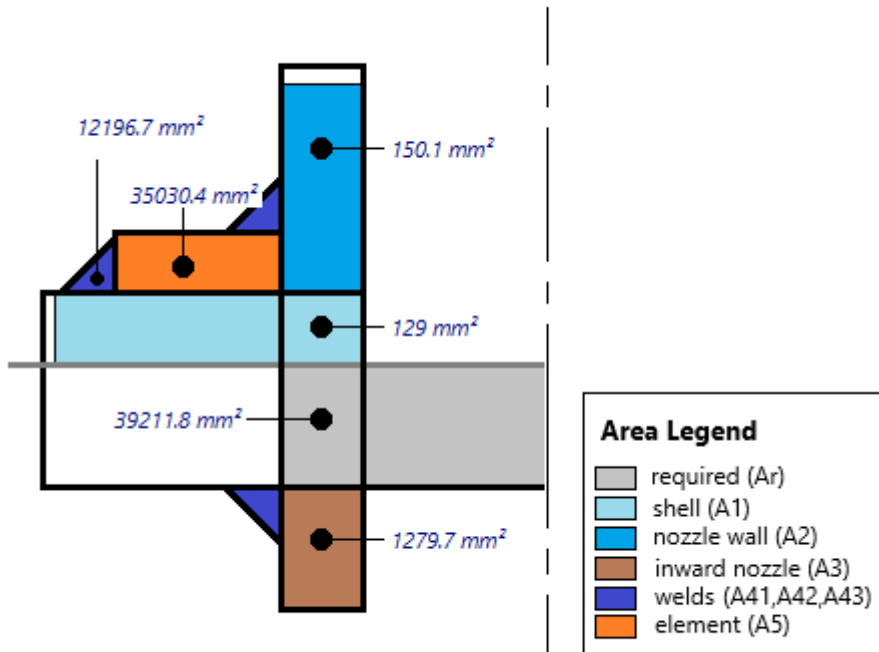


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	39211.758	2020.062	NA
Area in Shell	A1	128.965	35299.301	NA
Area in Nozzle Wall	A2	150.091	5576.271	NA
Area in Inward Nozzle	A3	1279.651	1279.651	NA
Area in Welds	A41+A42+A43	12196.669	12196.669	NA
Area in Element	A5	35030.434	35030.434	NA
TOTAL AREA AVAILABLE	Atot	48785.809	89382.328	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 774.4064 146.0000 mm
 Based on given Pad Diameter: 840.0000 106.0972 mm
 Based on the Estimated Diameter Limit: 1122.4126 48.7418 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (562 \cdot 69.8 \cdot 1 + 2 \cdot 19 \cdot 69.8 \cdot 1 \cdot (1 - 1))$
 $= 39211.758 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 562(1 \cdot 70 - 1 \cdot 69.8) - 2 \cdot 19$
 $(1 \cdot 70 - 1 \cdot 69.8) \cdot (1 - 1)$
 $= 128.965 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 175) \cdot (19 - 18.6) \cdot 1$

= 150.091 mm²

Area Available in Inward Nozzle [A3]:

$$\begin{aligned} &= 2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2 \\ &= 2 \cdot 16 \cdot (40) \cdot 1 \\ &= 1279.651 \text{ mm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr3 + (W_i\text{-can}/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 + 110^2 \cdot 1 \\ &= 12196.669 \text{ mm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tl_{wp}, t_e)) \cdot fr4 \\ &= (840 - 600) \cdot 146 \cdot 1 \\ &= 35030.434 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 21.5711 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 72.7705 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 72.7705 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.1185 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.3, \max(72.8, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(21.6, 11.3) \\ &= 21.5711 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 22.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 22, tr = 13.7, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.72, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-1 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 73, tr = 51, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.73, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 22, tr = 13.7, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.72, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
Min Metal Temp. at Required thickness (UCS 66.1) -20 °C

Gov. MDMT of the Nozzle : -1 °C
Gov. MDMT of the Reinforcement Pad : 10 °C
Gov. MDMT of the nozzle to shell joint welded assembly : 10 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6.41/10.20 = 0.628$

Weld Size Calculations, Description: U1600

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 * \text{TminPad}$	$77.7700 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (39212 - 129 + 2 \cdot 19 \cdot 1 \cdot (1 \cdot 70 - 69.8)) \cdot 101)$
 $= 3948984.75 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (150 + 35030 + 12197 - 0 \cdot 1) \cdot 101$
 $= 4785996.00 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (150 + 1280 + 100 + (2659)) \cdot 101$
 $= 423167.31 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (150 + 1280 + 12197 + 35030 + (2659)) \cdot 101$
 $= 5183902.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 600 \cdot 10 \cdot 0.49 \cdot 101$
 $= 466393. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 840 \cdot 110 \cdot 0.49 \cdot 101$
 $= 7182447. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 290) \cdot (22 - 3) \cdot 0.7 \cdot 101$
 $= 1225835. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$

$$= (3.14/2) \cdot 600 \cdot 146 \cdot 0.74 \cdot 101$$
$$= 10283481. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$
$$= (3.14/2.0) \cdot 600 \cdot (73 - 3) \cdot 0.74 \cdot 101$$
$$= 4931782. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (7182447 + 1225835) = 8408282 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (466393 + 10283481 + 4931782 + 0) = 15681656 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$
$$= (7182447 + 4931782 + 0) = 12114229 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 8408282 N , must exceed W = 3948984 N or W1 = 4785996 N
Path 2-2 = 15681656N , must exceed W = 3948984 N or W2 = 423167 N
Path 3-3 = 12114229N , must exceed W = 3948984 N or W3 = 5183902 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.43 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 20.7216 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 2573.0000 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.806 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K4150

From: 30

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	4364.00	mm
Head Finished (Minimum) Thickness	t	73.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
-------------------------------	----	-----------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

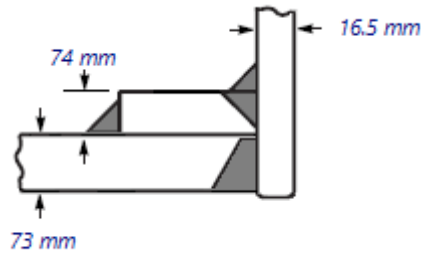
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	800.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	74.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	203.8000	mm
Thickness of Pad	te	74.0000	mm
Weld leg size between Pad and Shell	Wp	56.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	74.0000	mm
Reinforcing Pad Width		60.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K4150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	16.500 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
 $= 69.7705 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.41 \cdot 28.4 / (88.9 \cdot 1 - 0.6 \cdot 6.41)$
 $= 2.1407 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.7492 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	244.0656 mm
Parallel to Vessel Wall	Rn+tn+t	122.0328 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		107.7500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.88, 1)$
 $= 0.880$

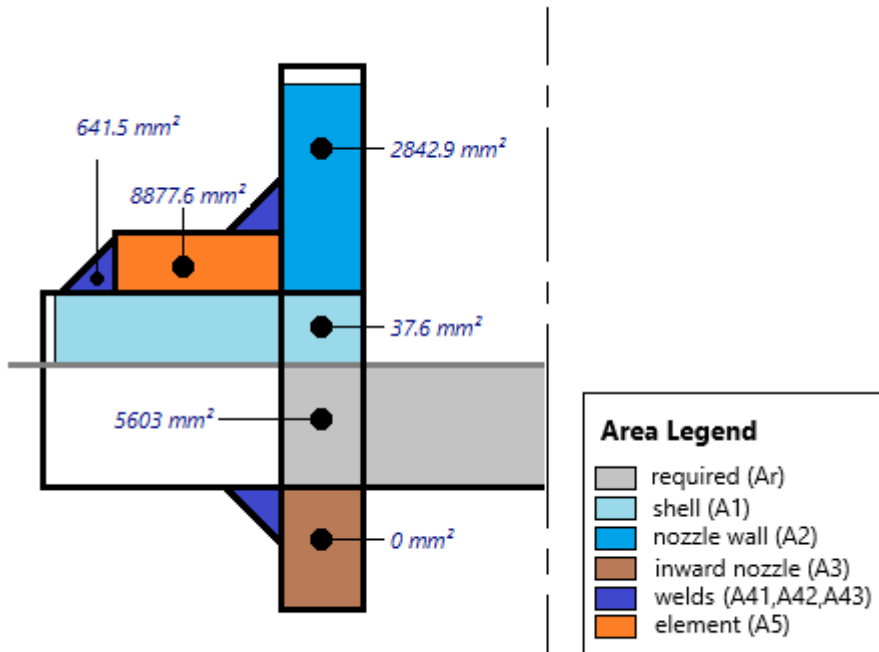


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5603.042	288.651	NA
Area in Shell	A1	37.579	10285.906	NA
Area in Nozzle Wall	A2	2842.888	3191.135	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	641.514	641.514	NA
Area in Element	A5	8877.577	8877.577	NA
TOTAL AREA AVAILABLE	Atot	12399.557	22996.133	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 47.48 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 111.9301 74.0000 mm
 Based on given Pad Diameter: 203.8000 17.3469 mm
 Based on Shell or Nozzle Thickness: 239.8582 16.5000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (77.1 \cdot 69.8 \cdot 1 + 2 \cdot 13.5 \cdot 69.8 \cdot 1 \cdot (1 - 0.88))$
 $= 5603.042 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 167(1 \cdot 70 - 1 \cdot 69.8) - 2 \cdot 13.5$
 $(1 \cdot 70 - 1 \cdot 69.8) \cdot (1 - 0.88)$
 $= 37.579 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 108) \cdot (13.5 - 2.14) \cdot 0.88 / \sin(49.3)$



= 2842.888 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (100) • 0.88 + (0) • 0.88 + 554² • 1
= 641.514 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (234 - 114) • 74 • 1
= 8877.577 mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 16.5, tr = 1.59, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.12, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 16.5, tr = 1.59, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.12, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 73, tr = 51, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.73, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.5, tr = 1.59, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.12, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: 10 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	10 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 6.41/10.20 = 0.628

Weld Size Calculations, Description: K4150

Intermediate Calc. for nozzle/shell Welds	Tmin	13.5000	mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	39.5920 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (5603 - 37.6 + 2 \cdot 13.5 \cdot 0.88 \cdot (1 \cdot 70 - 69.8)) \cdot 101) \\ &= 562768.12 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (2843 + 8878 + 642 - 0 \cdot 0.88) \cdot 101 \\ &= 1248794.62 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (2843 + 0 + 88 + (1663)) \cdot 101 \\ &= 464081.59 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (2843 + 0 + 642 + 8878 + (1663)) \cdot 101 \\ &= 1416801.25 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 88.9 \\ &= 77792. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.14/2.0) \cdot 204 \cdot 56 \cdot 0.49 \cdot 101 \\ &= 887141. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 47.7) \cdot (16.5 - 3) \cdot 0.7 \cdot 88.9 \\ &= 125859. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.14/2) \cdot 114 \cdot 74 \cdot 0.74 \cdot 101 \\ &= 987697. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 114 \cdot (73 - 3) \cdot 0.74 \cdot 101 \\ &= 934564. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (887141 + 125859) = 1013000 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\ &= (77792 + 987697 + 934564 + 0) = 2000053 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$



= (887141 + 934564 + 0) = 1821704 N

Summary of Failure Path Calculations:

Path 1-1 = 1013000 N , must exceed W = 562768 N or W1 = 1248794 N
Path 2-2 = 2000053 N , must exceed W = 562768 N or W2 = 464081 N
Path 3-3 = 1821704 N , must exceed W = 562768 N or W3 = 1416801 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.43 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K4150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.500 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
= 69.7705 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.41 \cdot 28.4 / (88.9 \cdot 1 - 0.6 \cdot 6.41)$
= 2.1407 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.7492 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	223.8000	mm
Parallel to Vessel Wall	Rn+tn+t	111.9000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		107.7500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4150.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.1407	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 72.7705	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 72.7705	mm
Wall Thickness, shell/head, external pressure	trb2	= 4.1185	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(72.8, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:



= max(ta, tb)
= max(5.14, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.5000 mm -> OK

Weld Size Calculations, Description: K4150

Intermediate Calc. for nozzle/shell Welds	Tmin	13.5000	mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	39.5920 = 0.7 * Wp mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 40.7393 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 939.8397 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: K180

From: 40

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	4300.00	mm
Design Length of Section	L	8154.6675	mm
Shell Finished (Minimum) Thickness	t	146.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2500.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

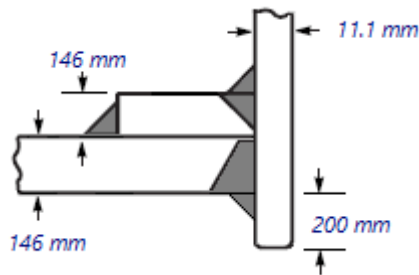
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	146.0000	mm
Inside Projection	h	200.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	128.9000	mm
Thickness of Pad	te	146.0000	mm
Weld leg size between Pad and Shell	Wp	110.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	146.0000	mm
Reinforcing Pad Width		20.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K180

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 66.650 mm.
 Actual Thickness Used in Calculation 11.125 mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.41 \cdot 2153 / (101 \cdot 1 - 0.6 \cdot 6.41)$
 $= 142.0327$ mm

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.41 \cdot 36.3 / (88.9 \cdot 1 - 0.6 \cdot 6.41)$
 $= 2.7380$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5125 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	374.9000	mm
Parallel to Vessel Wall	$R_n + t_n + t$	187.4500	mm
Normal to Vessel Wall (Thickness Limit), pad side T_{lp}		54.9540	mm
Normal to Vessel Wall, Inward		12.8130	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.88, 1)$
 $= 0.880$

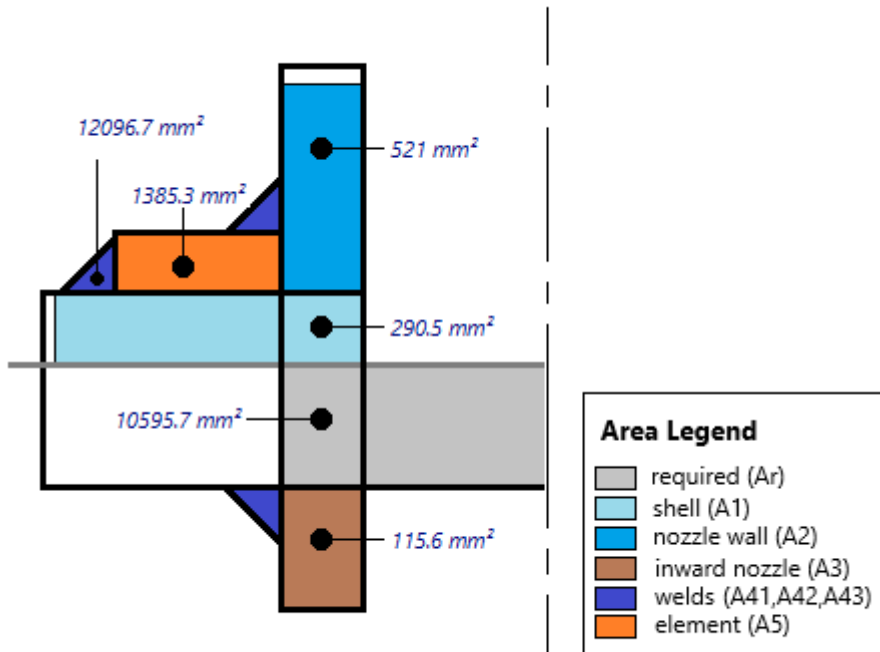


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10595.690	755.787	NA
Area in Shell	A1	290.465	36858.098	NA
Area in Nozzle Wall	A2	521.018	736.255	NA
Area in Inward Nozzle	A3	115.572	115.572	NA
Area in Welds	A41+A42+A43	12096.695	12096.695	NA
Area in Element	A5	1385.262	1385.262	NA
TOTAL AREA AVAILABLE	Atot	14409.013	51191.883	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 88.9000 146.0000 mm
 Based on given Pad Diameter: 128.9000 0.0000 mm
 Based on Shell or Nozzle Thickness: 88.9000 11.1252 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (72.6 \cdot 142 \cdot 1 + 2 \cdot 8.13 \cdot 142 \cdot 1 \cdot (1 - 0.88))$
 $= 10595.690 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 302(1 \cdot 143 - 1 \cdot 142) - 2 \cdot 8.13$
 $(1 \cdot 143 - 1 \cdot 142) \cdot (1 - 0.88)$
 $= 290.465 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 55) \cdot (8.13 - 2.74) \cdot 0.88$

= 521.018 mm²

Area Available in Inward Nozzle [A3]:

$$\begin{aligned} &= 2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2 \\ &= 2 \cdot 5.13 \cdot (12.8) \cdot 0.88 \\ &= 115.572 \text{ mm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= (W_o^2 - Ar \text{ Lost}) \cdot Fr3 + ((W_i - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4 \\ &= (0) \cdot 0.88 + (0) \cdot 0.88 + 110^2 \cdot 1 \\ &= 12096.695 \text{ mm}^2 \end{aligned}$$

Area Available in Element [A5]:

Pad Thickness is limited by 30 degree angle to 34.6410 mm

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (129 - 88.9) \cdot 34.6 \cdot 1 \\ &= 1385.262 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.7380 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 145.0327 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 145.0327 mm
Wall Thickness, shell/head, external pressure	trb2 = 5.2054 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.2054 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[7.8, \max(145, 5.21)] \\ &= 7.8000 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(5.74, 7.8) \\ &= 7.8000 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 2.04, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.73, tr = 2.04, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.
This component must be impact tested.

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B



Govrn. thk, $t_g = 9.73$, $t_r = 2.04$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6.41/10.20 = 0.628$

Weld Size Calculations, Description: K180

Intermediate Calc. for nozzle/shell Welds T_{min} 8.1252 mm
Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6876 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$77.7700 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (10596 - 290 + 2 \cdot 8.13 \cdot 0.88 \cdot (1 \cdot 143 - 142)) \cdot 101)$
 $= 1042420.69$ N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (521 + 1385 + 12097 - 0 \cdot 0.88) \cdot 101$
 $= 1414566.38$ N

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (521 + 116 + 88 + (2045)) \cdot 101$
 $= 279765.84$ N

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (521 + 116 + 12097 + 1385 + (2045)) \cdot 101$
 $= 1632810.12$ N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 88.9$
 $= 60825.$ N

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 129 \cdot 110 \cdot 0.49 \cdot 101$
 $= 1102164.$ N

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (11.1 - 3) \cdot 0.7 \cdot 88.9$
 $= 64149.$ N

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.14/2) \cdot 88.9 \cdot 146 \cdot 0.74 \cdot 101$
= 1523669. N

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
= $(3.14/2.0) \cdot 88.9 \cdot (146 - 3) \cdot 0.74 \cdot 101$
= 1492768. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SPEW + SNW) = (1102163 + 64149) = 1166312 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (60825 + 1523669 + 1492768 + 0) = 3077262 N

Failure Path [Path 3-3]:
= (Spew + Tngw + Sinw)
= (1102163 + 1492768 + 0) = 2594932 N

Summary of Failure Path Calculations:

Path 1-1 = 1166312 N , must exceed W = 1042420 N or W1 = 1414566 N
Path 2-2 = 3077262 N , must exceed W = 1042420 N or W2 = 279765 N
Path 3-3 = 2594932 N , must exceed W = 1042420 N or W3 = 1632810 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.45 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.4595 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 646.0000 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 187.45 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: K280
From: 50

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	4300.00	mm
Design Length of Section	L	8154.6675	mm
Shell Finished (Minimum) Thickness	t	146.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5000.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

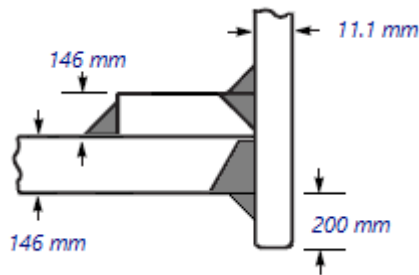
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	146.0000	mm
Inside Projection	h	200.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	128.9000	mm
Thickness of Pad	te	146.0000	mm
Weld leg size between Pad and Shell	Wp	110.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	146.0000	mm
Reinforcing Pad Width		20.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K280

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	66.650 mm.
Actual Thickness Used in Calculation	11.125 mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.41 \cdot 2153 / (101 \cdot 1 - 0.6 \cdot 6.41)$
 $= 142.0327 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.41 \cdot 36.3 / (88.9 \cdot 1 - 0.6 \cdot 6.41)$
 $= 2.7380 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5125 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	374.9000 mm
Parallel to Vessel Wall	$R_n + t_n + t$	187.4500 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		54.9540 mm
Normal to Vessel Wall, Inward		12.8130 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.88, 1)$
 $= 0.880$

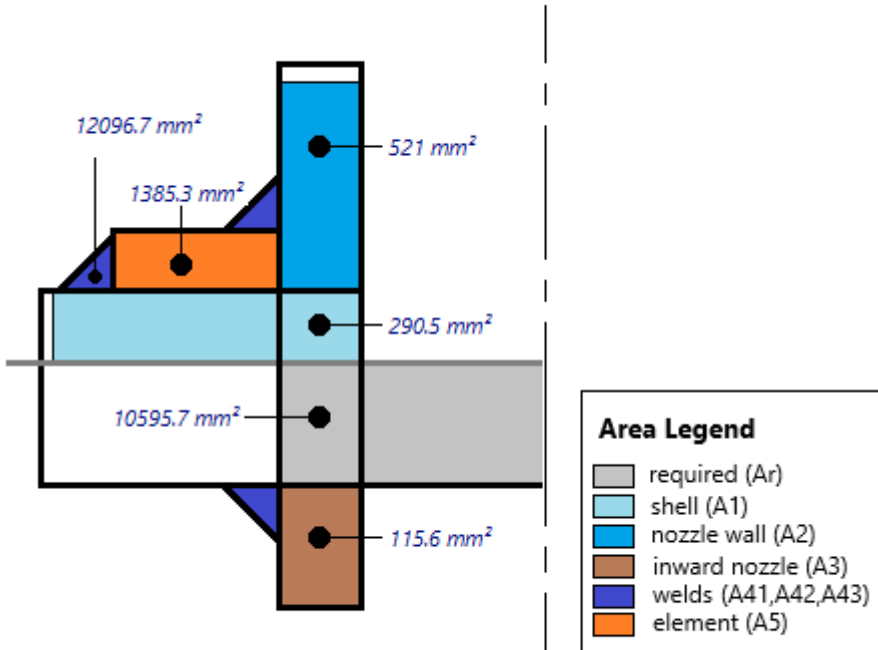


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10595.690	755.787	NA
Area in Shell	A1	290.465	36858.098	NA
Area in Nozzle Wall	A2	521.018	736.255	NA
Area in Inward Nozzle	A3	115.572	115.572	NA
Area in Welds	A41+A42+A43	12096.695	12096.695	NA
Area in Element	A5	1385.262	1385.262	NA
TOTAL AREA AVAILABLE	Atot	14409.013	51191.883	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 88.9000 146.0000 mm
 Based on given Pad Diameter: 128.9000 0.0000 mm
 Based on Shell or Nozzle Thickness: 88.9000 11.1252 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (72.6 \cdot 142 \cdot 1 + 2 \cdot 8.13 \cdot 142 \cdot 1 \cdot (1 - 0.88))$
 $= 10595.690 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 302(1 \cdot 143 - 1 \cdot 142) - 2 \cdot 8.13$
 $(1 \cdot 143 - 1 \cdot 142) \cdot (1 - 0.88)$
 $= 290.465 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 55) \cdot (8.13 - 2.74) \cdot 0.88$



= 521.018 mm²

Area Available in Inward Nozzle [A3]:

$$\begin{aligned} &= 2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2 \\ &= 2 \cdot 5.13 \cdot (12.8) \cdot 0.88 \\ &= 115.572 \text{ mm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= (W_o^2 - Ar \text{ Lost}) \cdot Fr3 + ((W_i - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4 \\ &= (0) \cdot 0.88 + (0) \cdot 0.88 + 110^2 \cdot 1 \\ &= 12096.695 \text{ mm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &\text{Pad Thickness is limited by 30 degree angle to } 34.6410 \text{ mm} \\ &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (129 - 88.9) \cdot 34.6 \cdot 1 \\ &= 1385.262 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.7380 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 145.0327 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 145.0327 mm
Wall Thickness, shell/head, external pressure	trb2 = 5.2054 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.2054 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[7.8, \max(145, 5.21)] \\ &= 7.8000 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(5.74, 7.8) \\ &= 7.8000 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 2.04, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.73, tr = 2.04, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.73, tr = 2.04, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.



This component must be impact tested.

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 2.04, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.3, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 6.41/10.20 = 0.628

Weld Size Calculations, Description: K280

Intermediate Calc. for nozzle/shell Welds Tmin 8.1252 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6876 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	77.7700 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (10596 - 290 + 2 • 8.13 • 0.88 •
(1 • 143 - 142))101)
= 1042420.69 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (521 + 1385 + 12097 - 0 • 0.88) • 101
= 1414566.38 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (521 + 116 + 88 + (2045)) • 101
= 279765.84 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (521 + 116 + 12097 + 1385 + (2045)) • 101
= 1632810.12 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 88.9 • 10 • 0.49 • 88.9
= 60825. N

Shear, Pad Element Weld [Spew]:
= (π/2) • DP • WP • 0.49 • SEW
= (3.14/2.0) • 129 • 110 • 0.49 • 101
= 1102164. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn



$$= (3.14 \cdot 40.4) \cdot (11.1 - 3) \cdot 0.7 \cdot 88.9$$
$$= 64149. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$
$$= (3.14/2) \cdot 88.9 \cdot 146 \cdot 0.74 \cdot 101$$
$$= 1523669. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$
$$= (3.14/2.0) \cdot 88.9 \cdot (146 - 3) \cdot 0.74 \cdot 101$$
$$= 1492768. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1102163 + 64149) = 1166312 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (60825 + 1523669 + 1492768 + 0) = 3077262 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$
$$= (1102163 + 1492768 + 0) = 2594932 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1166312 N	, must exceed W = 1042420 N	or W1 = 1414566 N
Path 2-2 = 3077262 N	, must exceed W = 1042420 N	or W2 = 279765 N
Path 3-3 = 2594932 N	, must exceed W = 1042420 N	or W3 = 1632810 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.45 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.4595 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 646.0000 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: K4280

From: 60

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	4300.00	mm
Design Length of Section	L	8154.6675	mm
Shell Finished (Minimum) Thickness	t	146.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		6300.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

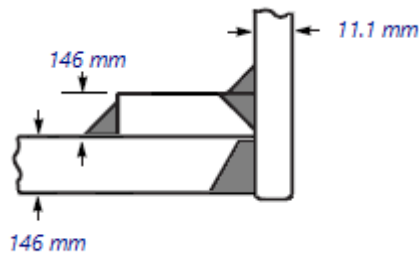
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	146.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	288.9000	mm
Thickness of Pad	te	146.0000	mm
Weld leg size between Pad and Shell	Wp	110.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	146.0000	mm
Reinforcing Pad Width		100.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K4280

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	88.900 mm.
Actual Thickness Used in Calculation	11.125 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.41 \cdot 2153 / (101 \cdot 1 - 0.6 \cdot 6.41)$
 $= 142.0327 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.41 \cdot 44.5 / (88.9 \cdot 1 + 0.4 \cdot 6.41)$
 $= 3.1156 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4346 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	374.9000 mm
Parallel to Vessel Wall	Rn+tn+t	187.4500 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		166.3130 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.88, 1)$
 $= 0.880$

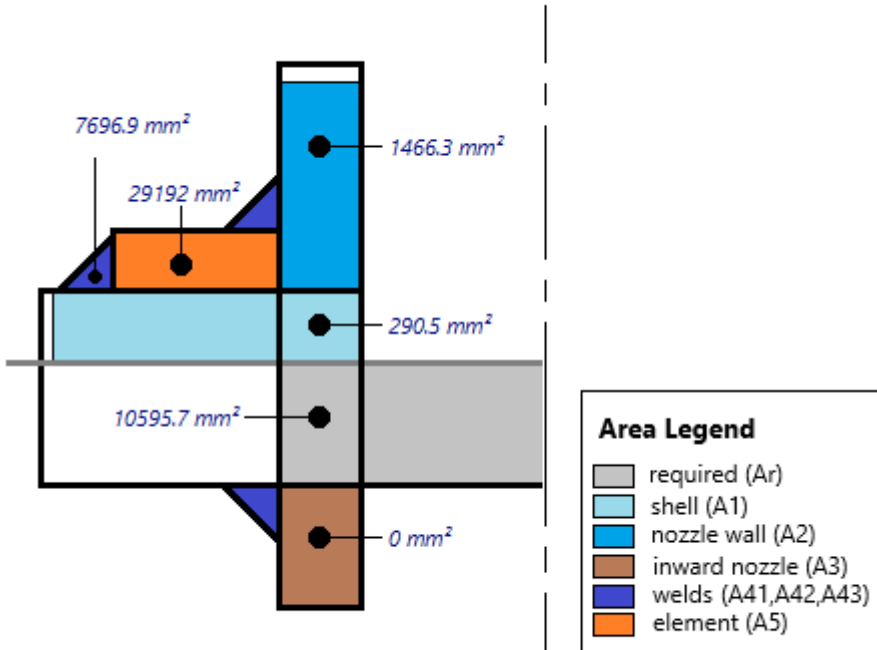


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10595.690	755.787	NA
Area in Shell	A1	290.465	36858.098	NA
Area in Nozzle Wall	A2	1466.287	2251.004	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	7696.919	7696.919	NA
Area in Element	A5	29192.031	29192.031	NA
TOTAL AREA AVAILABLE	Atot	38645.703	75998.047	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 96.7242 146.0000 mm
 Based on given Pad Diameter: 288.9000 5.7117 mm
 Based on Shell or Nozzle Thickness: 191.5795 11.1252 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (72.6 \cdot 142 \cdot 1 + 2 \cdot 8.13 \cdot 142 \cdot 1 \cdot (1 - 0.88))$
 $= 10595.690 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 302(1 \cdot 143 - 1 \cdot 142) - 2 \cdot 8.13$
 $(1 \cdot 143 - 1 \cdot 142) \cdot (1 - 0.88)$
 $= 290.465 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 166) \cdot (8.13 - 3.12) \cdot 0.88$



= 1466.287 mm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (100) • 0.88 + (0) • 0.88 + 7611² • 1
= 7696.919 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (289 - 88.9) • 146 • 1
= 29192.031 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.1156 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 145.0327 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 145.0327 mm
Wall Thickness, shell/head, external pressure	trb2 = 5.2054 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.2054 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(145, 5.21)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(6.12, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 0.875 * 11.125 = 9.735 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 2.36, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.35, Temp. Reduction = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.73, tr = 2.36, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.35, Temp. Reduction = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.73, tr = 2.36, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.35, Temp. Reduction = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 2.36, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.35, Temp. Reduction = 77 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6.41/10.20 = 0.628$

Weld Size Calculations, Description: K4280

Intermediate Calc. for nozzle/shell Welds Tmin 8.1252 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6876 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$77.7700 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (10596 - 290 + 2 \cdot 8.13 \cdot 0.88 \cdot (1 \cdot 143 - 142)) \cdot 101)$
 $= 1042420.69 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1466 + 29192 + 7697 - 0 \cdot 0.88) \cdot 101$
 $= 3874607.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1466 + 0 + 88 + (2045)) \cdot 101$
 $= 363580.97 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1466 + 0 + 7697 + 29192 + (2045)) \cdot 101$
 $= 4081176.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 88.9$
 $= 60825. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 289 \cdot 110 \cdot 0.49 \cdot 101$
 $= 2470248. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (11.1 - 3) \cdot 0.7 \cdot 88.9$
 $= 64149. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 88.9 \cdot 146 \cdot 0.74 \cdot 101$



= 1523669. N

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 88.9 \cdot (146 - 3) \cdot 0.74 \cdot 101 \\ &= 1492768. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (2470248 + 64149) = 2534397 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (60825 + 1523669 + 1492768 + 0) = 3077262 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (2470248 + 1492768 + 0) = 3963017 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 2534397 N , must exceed W = 1042420 N or W1 = 3874607 N

Path 2-2 = 3077262 N , must exceed W = 1042420 N or W2 = 363580 N

Path 3-3 = 3963017 N , must exceed W = 1042420 N or W3 = 4081176 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.45 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 0.4595 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 346.4595 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Input, Nozzle Desc: M1600

From: 70

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	4364.00	mm
Head Finished (Minimum) Thickness	t	73.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	101.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

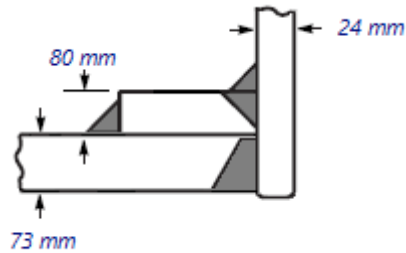
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	73.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1200.0000	mm
Thickness of Pad	te	80.0000	mm
Weld leg size between Pad and Shell	Wp	56.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	80.0000	mm
Reinforcing Pad Width		300.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M1600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	600.000 mm.
Actual Thickness Used in Calculation	24.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
 $= 69.7705 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.41 \cdot 300 / (101 \cdot 1 + 0.4 \cdot 6.41)$
 $= 18.5711 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.7580 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1116.0001 mm
Parallel to Vessel Wall, opening length	d	558.0001 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		132.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

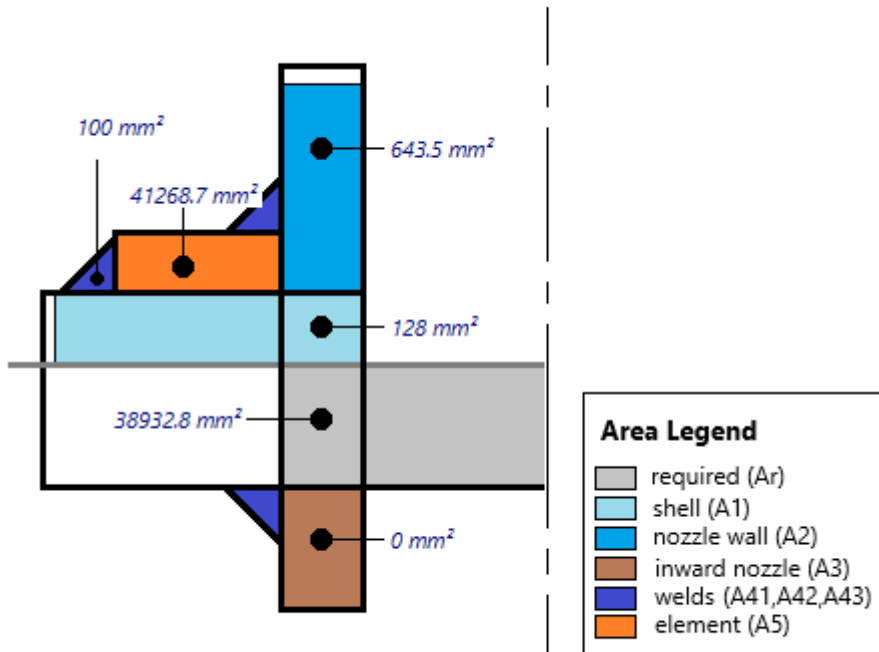


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	38932.754	2005.689	NA
Area in Shell	A1	128.047	35047.984	NA
Area in Nozzle Wall	A2	643.496	5097.731	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	99.973	99.973	NA
Area in Element	A5	41268.738	41268.738	NA
TOTAL AREA AVAILABLE	Atot	42140.254	81514.430	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1075.8954 80.0000 mm
 Based on given Pad Diameter: 1200.0000 73.7822 mm
 Based on the Estimated Diameter Limit: 1114.4126 74.0099 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (558 \cdot 69.8 \cdot 1 + 2 \cdot 21 \cdot 69.8 \cdot 1 \cdot (1 - 1))$
 $= 38932.754 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 558(1 \cdot 70 - 1 \cdot 69.8) - 2 \cdot 21$
 $(1 \cdot 70 - 1 \cdot 69.8) \cdot (1 - 1)$
 $= 128.047 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 132) \cdot (21 - 18.6) \cdot 1$

= 643.496 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$
= 99.973 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(1116 - 600) \cdot 80 \cdot 1$
= 41268.738 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 21.5711$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
= 21.5711 mm

Available Nozzle Neck Thickness = 24.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 24$, $t_r = 13.7$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.65$, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-2 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 24$, $t_r = 13.7$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.65$, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-21 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 73$, $t_r = 51$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.73$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 24$, $t_r = 13.7$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.65$, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-21 °C

Gov. MDMT of the Nozzle	:	-2 °C
Gov. MDMT of the Reinforcement Pad	:	10 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	10 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :



Design Pressure/Ambient Rating = 6.41/10.20 = 0.628

Weld Size Calculations, Description: M1600

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	39.5920 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (38933 - 128 + 2 • 21 • 1 •
(1 • 70 - 69.8))101)
= 3920985.25 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (643 + 41269 + 100 - 0 • 1) • 101
= 4244031.00 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (643 + 0 + 100 + (2939)) • 101
= 372019.22 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (643 + 0 + 100 + 41269 + (2939)) • 101
= 4540945.50 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 600 • 10 • 0.49 • 101
= 466393. N

Shear, Pad Element Weld [Spew]:
= (π/2) • DP • WP • 0.49 • SEW
= (3.14/2.0) • 1200 • 56 • 0.49 • 101
= 5223598. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 290) • (24 - 3) • 0.7 • 101
= 1350206. N

Tension, Pad Groove Weld [Tpgw]:
= (π/2) • Dlo • Wgpn • 0.74 • Seg
= (3.14/2) • 600 • 80 • 0.74 • 101
= 5634784. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 600 • (73 - 3) • 0.74 • 101
= 4931782. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SPEW + SNW) = (5223597 + 1350206) = 6573804 N



Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (466393 + 5634784 + 4931782 + 0) = 11032959 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$
$$= (5223597 + 4931782 + 0) = 10155380 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 6573804 N , must exceed W = 3920985 N or W1 = 4244031 N
Path 2-2 = 11032959N , must exceed W = 3920985 N or W2 = 372019 N
Path 3-3 = 10155380N , must exceed W = 3920985 N or W3 = 4540945 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.43 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 20.7216 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 493.7216 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 * t_{nom} / R_f * (1 - R_f / R_o)$) 4.167 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K350

From: 70

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	4364.00	mm
Head Finished (Minimum) Thickness	t	73.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
-------------------------------	----	-----------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

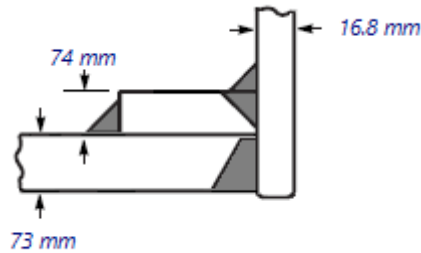
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	800.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	74.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	184.3280	mm
Thickness of Pad	te	74.0000	mm
Weld leg size between Pad and Shell	Wp	56.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	74.0000	mm
Reinforcing Pad Width		49.9640	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K350

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	16.800 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
 $= 69.7705 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.41 \cdot 28.4 / (88.9 \cdot 1 - 0.6 \cdot 6.41)$
 $= 2.1407 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.7532 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	244.6656 mm
Parallel to Vessel Wall	Rn+tn+t	122.3328 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		108.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.88, 1)$
 $= 0.880$

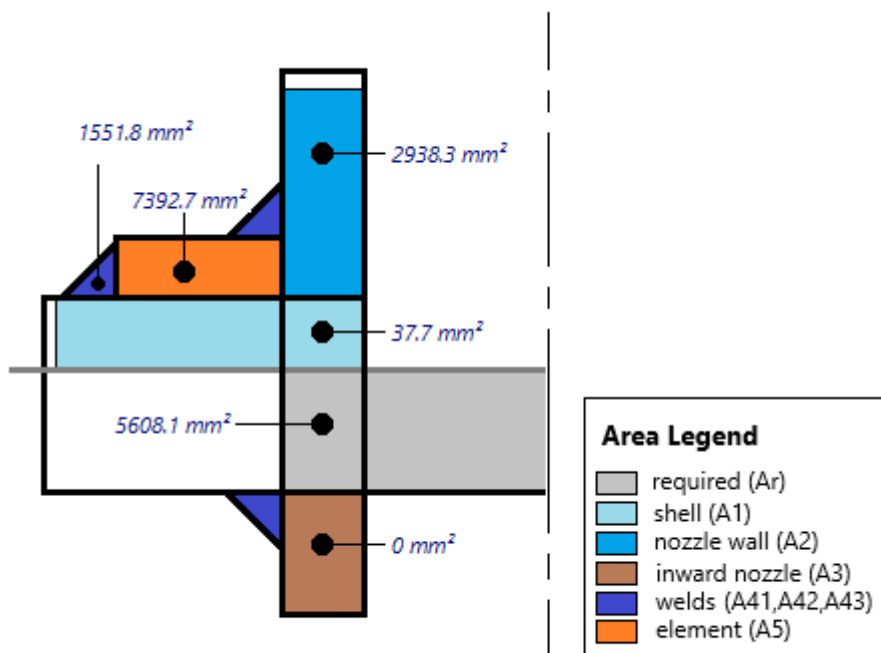


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5608.067	288.909	NA
Area in Shell	A1	37.700	10319.070	NA
Area in Nozzle Wall	A2	2938.279	3287.929	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1551.795	1551.795	NA
Area in Element	A5	7392.653	7392.653	NA
TOTAL AREA AVAILABLE	Atot	11920.429	22551.447	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 47.48 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	99.0025	74.0000 mm
Based on given Pad Diameter:	184.3280	10.8137 mm
Based on Shell or Nozzle Thickness:	178.8336	16.8000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (77.1 \cdot 69.8 \cdot 1 + 2 \cdot 13.8 \cdot 69.8 \cdot 1 \cdot (1 - 0.88))$
 $= 5608.067 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 168(1 \cdot 70 - 1 \cdot 69.8) - 2 \cdot 13.8$
 $(1 \cdot 70 - 1 \cdot 69.8) \cdot (1 - 0.88)$
 $= 37.700 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 108) \cdot (13.8 - 2.14) \cdot 0.88 / \sin(49.3)$



= 2938.279 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (100) • 0.88 + (0) • 0.88 + 1464² • 1
= 1551.795 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (214 - 115) • 74 • 1
= 7392.653 mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 16.8, tr = 1.59, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.12, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.8, tr = 1.59, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.12, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 73, tr = 51, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.73, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 1.59, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.12, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	:	-104 °C
Gov. MDMT of the Reinforcement Pad	:	10 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	10 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 6.41/10.20 = 0.628

Weld Size Calculations, Description: K350

Intermediate Calc. for nozzle/shell Welds	Tmin	13.8000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	39.5920 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (5608 - 37.7 + 2 \cdot 13.8 \cdot 0.88 \cdot (1 \cdot 70 - 69.8)) \cdot 101) \\ &= 563275.75 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (2938 + 7393 + 1552 - 0 \cdot 0.88) \cdot 101 \\ &= 1200381.00 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (2938 + 0 + 88 + (1700)) \cdot 101 \\ &= 477451.44 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (2938 + 0 + 1552 + 7393 + (1700)) \cdot 101 \\ &= 1372121.25 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 115 \cdot 10 \cdot 0.49 \cdot 88.9 \\ &= 78349. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.14/2.0) \cdot 184 \cdot 56 \cdot 0.49 \cdot 101 \\ &= 802379. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 47.9) \cdot (16.8 - 3) \cdot 0.7 \cdot 88.9 \\ &= 129205. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.14/2) \cdot 115 \cdot 74 \cdot 0.74 \cdot 101 \\ &= 994769. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 115 \cdot (73 - 3) \cdot 0.74 \cdot 101 \\ &= 941255. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (802379 + 129205) = 931584 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\ &= (78349 + 994769 + 941255 + 0) = 2014373 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$



= (802379 + 941255 + 0) = 1743634 N

Summary of Failure Path Calculations:

Path 1-1 = 931584 N , must exceed W = 563275 N or W1 = 1200381 N
Path 2-2 = 2014373 N , must exceed W = 563275 N or W2 = 477451 N
Path 3-3 = 1743634 N , must exceed W = 563275 N or W3 = 1372121 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.43 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K350

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.800 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
= 69.7705 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.41 \cdot 28.4 / (88.9 \cdot 1 - 0.6 \cdot 6.41)$
= 2.1407 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.7532 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	224.4000 mm
Parallel to Vessel Wall	Rn+tn+t	112.2000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		108.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K350.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.1407 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 72.7705 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 72.7705 mm
Wall Thickness, shell/head, external pressure	trb2	= 4.1185 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(72.8, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:



= max(ta, tb)
= max(5.14, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Weld Size Calculations, Description: K350

Intermediate Calc. for nozzle/shell Welds	Tmin	13.8000	mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	39.5920 = 0.7 * Wp mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 41.0390 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 940.1394 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K580

From: 70

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	4364.00	mm
Head Finished (Minimum) Thickness	t	73.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
-------------------------------	----	-----------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

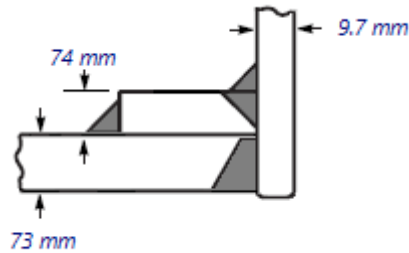
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	800.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	74.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	288.9000	mm
Thickness of Pad	te	74.0000	mm
Weld leg size between Pad and Shell	Wp	56.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	74.0000	mm
Reinforcing Pad Width		100.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K580

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	88.900 mm.
Actual Thickness Used in Calculation	9.735 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
 $= 69.7705 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.41 \cdot 44.5 / (88.9 \cdot 1 + 0.4 \cdot 6.41)$
 $= 3.1156 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.7834 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	255.8265 mm
Parallel to Vessel Wall	Rn+tn+t	127.9132 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		90.8364 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101 / 101)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.88, 1)$
 $= 0.880$

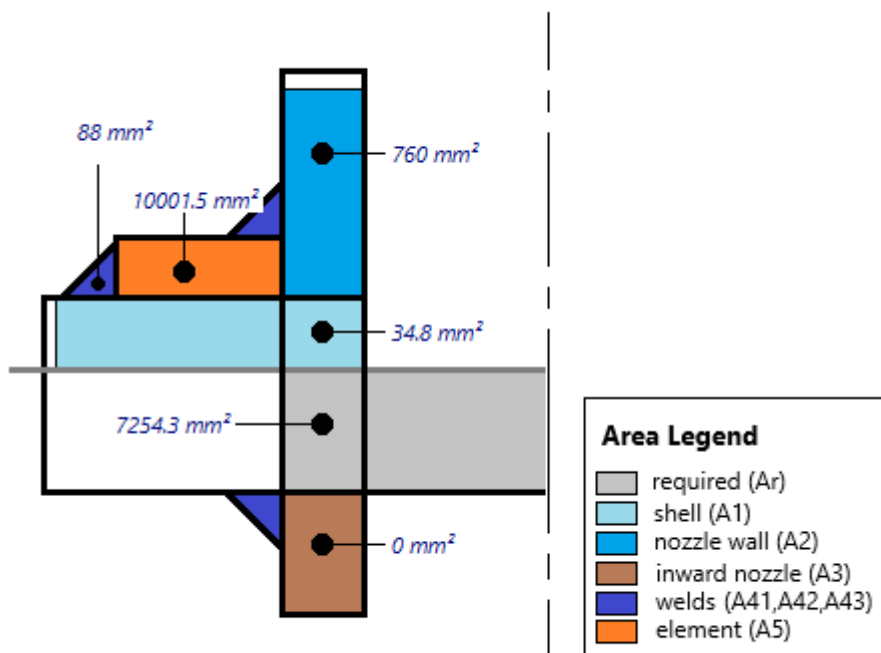


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	7254.338	373.720	NA
Area in Shell	A1	34.847	9538.035	NA
Area in Nozzle Wall	A2	759.987	1249.772	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	87.996	87.996	NA
Area in Element	A5	10001.471	10001.471	NA
TOTAL AREA AVAILABLE	Atot	10884.301	20877.273	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 47.47 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	175.0250	74.0000 mm
Based on given Pad Diameter:	288.9000	47.1422 mm
Based on the Estimated Diameter Limit:	254.2390	47.7024 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (102 \cdot 69.8 \cdot 1 + 2 \cdot 6.73 \cdot 69.8 \cdot 1 \cdot (1 - 0.88))$
 $= 7254.338 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 153(1 \cdot 70 - 1 \cdot 69.8) - 2 \cdot 6.73$
 $(1 \cdot 70 - 1 \cdot 69.8) \cdot (1 - 0.88)$
 $= 34.847 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 90.8) \cdot (6.73 - 3.12) \cdot 0.88 / \sin(49.6)$



= 759.987 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10^2 \cdot 0.88 + (0)^2 \cdot 0.88 + 0^2 \cdot 1$
= 87.996 mm²

Area Available in Element [A5]:
= $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$
= $(256 - 121) \cdot 74 \cdot 1$
= 10001.471 mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.73, tr = 2.36, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.35$, Temp. Reduction = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.73, tr = 2.36, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.35$, Temp. Reduction = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 73, tr = 51, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.73$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 2.36, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.35$, Temp. Reduction = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	10 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	10 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6.41 / 10.20 = 0.628$

Weld Size Calculations, Description: K580

Intermediate Calc. for nozzle/shell Welds	Tmin	6.7346 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$39.5920 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (7254 - 34.8 + 2 \cdot 6.73 \cdot 0.88 \cdot (1 \cdot 70 - 69.8)) \cdot 101)$$

$$= 729580.50 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - Can) \cdot 0.707) \cdot f_{r2} \cdot S_v$$

$$= (760 + 10001 + 88 - 0 \cdot 0.88) \cdot 101$$

$$= 1096000.88 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (760 + 0 + 88 + (830)) \cdot 101$$

$$= 169473.44 \text{ N}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (760 + 0 + 88 + 10001 + (830)) \cdot 101$$

$$= 1179812.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 121 \cdot 10 \cdot 0.49 \cdot 88.9$$

$$= 82538. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 289 \cdot 56 \cdot 0.49 \cdot 101$$

$$= 1257581. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 55.7) \cdot (9.73 - 3) \cdot 0.7 \cdot 88.9$$

$$= 73392. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 121 \cdot 74 \cdot 0.74 \cdot 101$$

$$= 1047948. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 121 \cdot (73 - 3) \cdot 0.74 \cdot 101$$

$$= 991572. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (1257581 + 73392) = 1330973 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (82538 + 1047947 + 991572 + 0) = 2122057 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

= (1257581 + 991572 + 0) = 2249153 N

Summary of Failure Path Calculations:

Path 1-1 = 1330973 N , must exceed W = 729580 N or W1 = 1096000 N
Path 2-2 = 2122057 N , must exceed W = 729580 N or W2 = 169473 N
Path 3-3 = 2249153 N , must exceed W = 729580 N or W3 = 1179812 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.43 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K580

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	88.900 mm.
Actual Thickness Used in Calculation	9.735 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
= 69.7705 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $6.41 \cdot 44.5 / (88.9 \cdot 1 + 0.4 \cdot 6.41)$
= 3.1156 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.7834 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	228.9000 mm
Parallel to Vessel Wall	Rn+tn+t	114.4500 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		90.8364 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

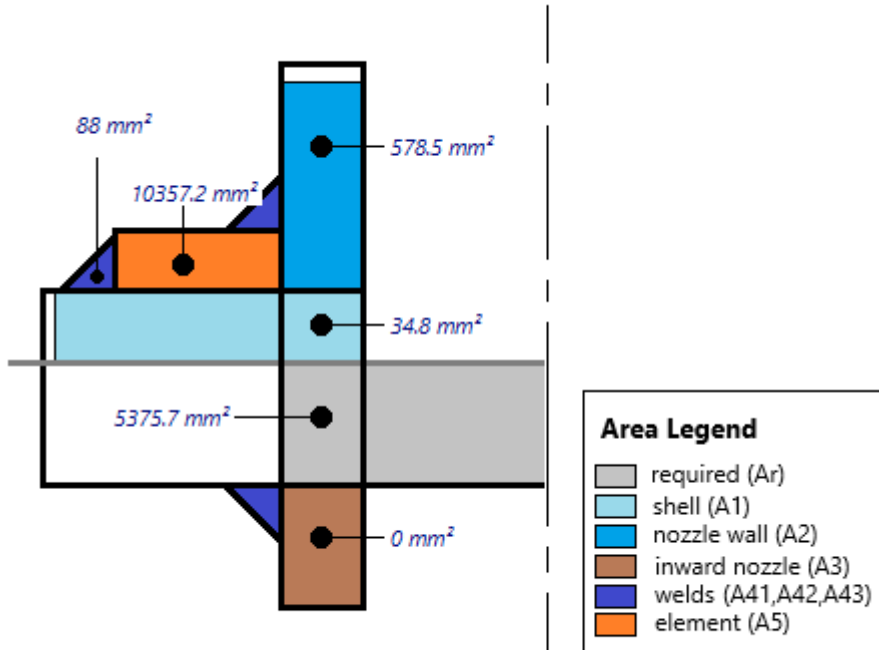


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5375.662	276.937	NA
Area in Shell	A1	34.847	9538.035	NA
Area in Nozzle Wall	A2	578.537	951.384	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	87.996	87.996	NA
Area in Element	A5	10357.171	10357.171	NA
TOTAL AREA AVAILABLE	Atot	11058.551	20934.588	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	152.0832	74.0000 mm
Based on given Pad Diameter:	288.9000	33.3968 mm
Based on the Estimated Diameter Limit:	227.3125	33.7799 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (75.4 \cdot 69.8 \cdot 1 + 2 \cdot 6.73 \cdot 69.8 \cdot 1 \cdot (1 - 0.88))$
 $= 5375.662 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 153(1 \cdot 70 - 1 \cdot 69.8) - 2 \cdot 6.73$
 $(1 \cdot 70 - 1 \cdot 69.8) \cdot (1 - 0.88)$
 $= 34.847 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 90.8) \cdot (6.73 - 3.12) \cdot 0.88$

= 578.537 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 0.88 + (0)^2 \cdot 0.88 + 0^2 \cdot 1$
 = 87.996 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(229 - 88.9) \cdot 74 \cdot 1$
 = 10357.171 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.1156 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 72.7705 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 72.7705 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.1185 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[7.8, max(72.8, 4.5)]
 = 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(6.12, 7.8)
 = 7.8000 mm

Available Nozzle Neck Thickness = 9.7346 mm -> OK

Weld Size Calculations, Description: K580

Intermediate Calc. for nozzle/shell Welds	Tmin	6.7346 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.7142 = $0.7 \cdot t_{min}$	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	39.5920 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = $\max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 = $\max(0, (5376 - 34.8 + 2 \cdot 6.73 \cdot 0.88 \cdot (1 \cdot 70 - 69.8)) \cdot 101)$
 = 539798.44 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = $(A2 + A5 + A4 - (W_i - Can/0.707)^2 \cdot fr_2) \cdot Sv$
 = $(579 + 10357 + 88 - 0 \cdot 0.88) \cdot 101$
 = 1113603.38 N

Weld Load [W2]:
 = $(A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$
 = $(579 + 0 + 88 + (830)) \cdot 101$
 = 151143.55 N

Weld Load [W3]:
 = $(A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$
 = $(579 + 0 + 88 + 10357 + (830)) \cdot 101$
 = 1197414.50 N

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 88.9 \\ &= 60825. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.14/2.0) \cdot 289 \cdot 56 \cdot 0.49 \cdot 101 \\ &= 1257581. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 41.1) \cdot (9.73 - 3) \cdot 0.7 \cdot 88.9 \\ &= 54085. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.14/2) \cdot 88.9 \cdot 74 \cdot 0.74 \cdot 101 \\ &= 772271. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 88.9 \cdot (73 - 3) \cdot 0.74 \cdot 101 \\ &= 730726. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1257581 + 54085) = 1311666 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (60825 + 772271 + 730726 + 0) = 1563821 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ &= (1257581 + 730726 + 0) = 1988306 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1311666 N	, must exceed W = 539798 N	or W1 = 1113603 N
Path 2-2 = 1563821 N	, must exceed W = 539798 N	or W2 = 151143 N
Path 3-3 = 1988306 N	, must exceed W = 539798 N	or W3 = 1197414 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.43 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 43.2902 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 942.3906 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N1400

From: 70

Pressure for Reinforcement Calculations	P	6.4110	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	4364.00	mm
Head Finished (Minimum) Thickness	t	73.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
-------------------------------	----	-----------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		60.00	deg
Diameter		400.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	
-----------------	--	--------	--

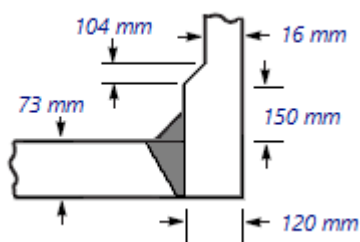
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	104.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	700.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		73.0000	mm
Inside Projection	h	500.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N1400

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	400.000	mm.
Actual Thickness Used in Calculation	16.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$
= 69.7705 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $6.41 \cdot 200 / (101 \cdot 1 + 0.4 \cdot 6.41)$
= 12.3807 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.7543 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 104/0.5773
= 180.1334 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

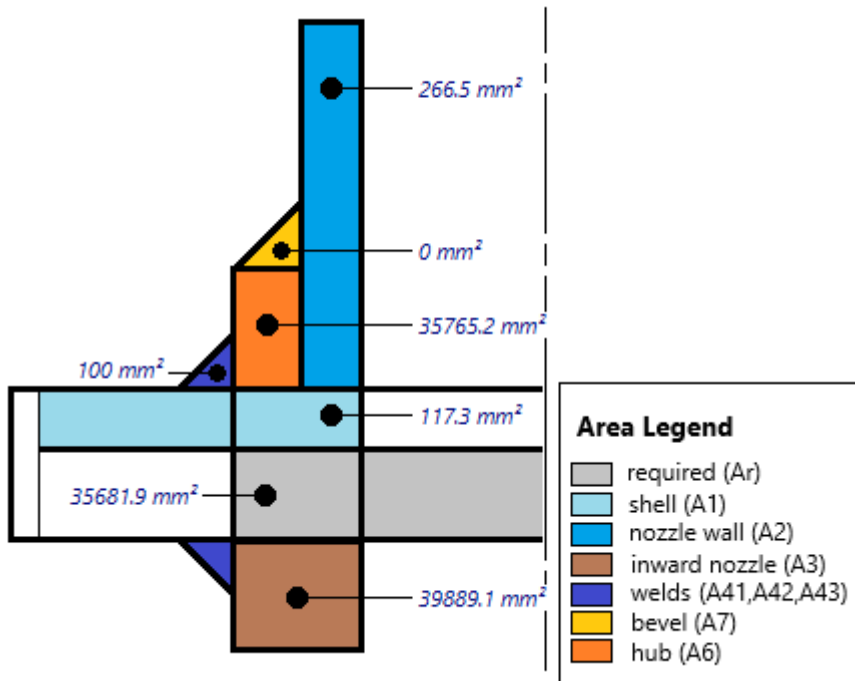
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1022.7087	mm
Parallel to Vessel Wall, opening length	d	511.3543	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	175.0000	mm
Normal to Vessel Wall, Inward		25.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 101/101)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 101/101)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	35681.918	1838.216	NA
Area in Shell	A1	117.331	32114.820	NA
Area in Nozzle Wall	A2	266.538	4840.006	NA
Area in Inward Nozzle	A3	39889.109	39889.109	NA
Area in Welds	A41+A42+A43	99.973	99.973	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	35765.242	35765.242	NA
TOTAL AREA AVAILABLE	Atot	76138.188	112709.148	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 47.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (511 \cdot 69.8 \cdot 1 + 2 \cdot 117 \cdot 69.8 \cdot 1 \cdot (1 - 1))$
 $= 35681.918 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 511(1 \cdot 70 - 1 \cdot 69.8) - 2 \cdot 117$
 $(1 \cdot 70 - 1 \cdot 69.8) \cdot (1 - 1)$
 $= 117.331 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 175)(13 - 12.4)/\sin(54.4)$
 $= 266.538 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 114 \cdot (175) \cdot 1$

= 39889.109 mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 99.973 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(175, 700, 150)) \cdot (120 - 16) \cdot 1$$

$$= 35765.242 \text{ mm}^2 \text{ Includes Bevel Area of } 4573.759 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 16, tr = 9.12, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.7, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 73, tr = 51, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.73, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 10 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 6.41/10.20 = 0.628

Weld Size Calculations, Description: N1400

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (35682 - 117 + 2 \cdot 117 \cdot 1 \cdot (1 \cdot 70 - 69.8)) \cdot 101)$$

$$= 3598121.50 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can/0.707)^2 \cdot fr2) \cdot Sv$$

$$= (36032 + 100 - 0 \cdot 1) \cdot 101$$

$$= 3649993.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (36032 + 39889 + 100 + (16376)) \cdot 101$$

$$= 9333790.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (36032 + 39889 + 100 + 0 + (16376)) \cdot 101$$

$$= 9333790.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 547 \cdot 10 \cdot 0.49 \cdot 101$$

$$= 425119. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 265) \cdot (120 - 3) \cdot 0.7 \cdot 101$$

$$= 6874634. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 547 \cdot (73 - 3) \cdot 0.74 \cdot 101$$

$$= 4495345. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (425119 + 6874634) = 7299753 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (425119 + 0 + 4495345 + 0) = 4920464 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (425119 + 4495345 + 0) = 4920464 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 7299753 N	, must exceed W = 3598121 N	or W1 = 3649993 N
Path 2-2 = 4920464 N	, must exceed W = 3598121 N	or W2 = 9333790 N
Path 3-3 = 4920464 N	, must exceed W = 3598121 N	or W3 = 9333790 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.43 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N1400

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	400.000 mm.
Actual Thickness Used in Calculation	16.000 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= 6.41 \cdot 2185 / (2 \cdot 101 \cdot 1 - 0.2 \cdot 6.41)$$

$$= 69.7705 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.41 \cdot 200 / (101 \cdot 1 + 0.4 \cdot 6.41)$
 $= 12.3807 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.7543 mm

Intermediate Hub Nozzle Calculations:

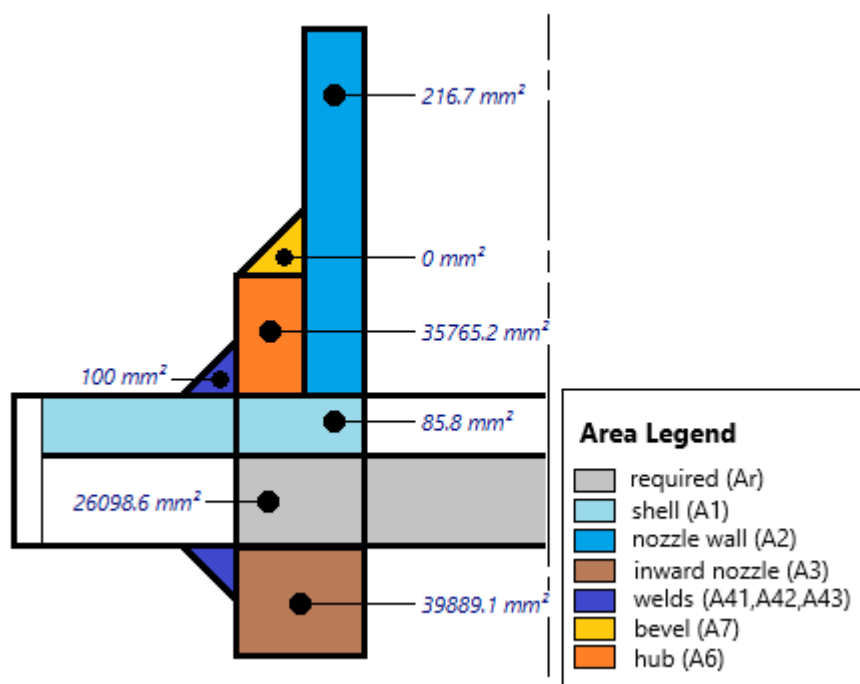
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 104 / 0.5773$
 $= 180.1334 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	748.0000	mm
Parallel to Vessel Wall, opening length	d	374.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	175.0000	mm
Normal to Vessel Wall, Inward		25.0000	mm



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26098.631	1344.517	NA
Area in Shell	A1	85.811	23487.412	NA
Area in Nozzle Wall	A2	216.695	3934.932	NA
Area in Inward Nozzle	A3	39889.109	39889.109	NA
Area in Welds	A41+A42+A43	99.973	99.973	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	35765.242	35765.242	NA
TOTAL AREA AVAILABLE	Atot	76056.828	103176.672	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (374 \cdot 69.8 \cdot 1 + 2 \cdot 117 \cdot 69.8 \cdot 1 \cdot (1 - 1))$
 $= 26098.631 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 374(1 \cdot 70 - 1 \cdot 69.8) - 2 \cdot 117$
 $(1 \cdot 70 - 1 \cdot 69.8) \cdot (1 - 1)$
 $= 85.811 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 175)(13 - 12.4)1$
 $= 216.695 \text{ mm}^2$

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 114 \cdot (175) \cdot 1$
 $= 39889.109 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 99.973 \text{ mm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$
 $= (2 \cdot \min(175, 700, 150)) \cdot (120 - 16) \cdot 1$
 $= 35765.242 \text{ mm}^2$ Includes Bevel Area of 4573.759 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 15.3807 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 72.7705 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 72.7705 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.1185 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.3, \max(72.8, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(15.4, 11.3)$
 $= 15.3807 \text{ mm}$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	77.4,	Allowable	:	101.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	221.4 MPa	Passed
Occasional	:	44.6,	Allowable	:	134.3 MPa	Passed
Shear	:	17.7,	Allowable	:	70.7 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N1400

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (26099 - 85.8 + 2 • 117 • 1 •
(1 • 70 - 69.8))101)
= 2633211.75 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (35982 + 100 - 0 • 1) • 101
= 3644958.00 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (35982 + 39889 + 100 + (16376)) • 101
= 9328756.00 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (35982 + 39889 + 100 + 0 + (16376)) • 101
= 9328756.00 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 400 • 10 • 0.49 • 101
= 310928. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 194) • (120 - 3) • 0.7 • 101
= 5028046. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 400 • (73 - 3) • 0.74 • 101
= 3287855. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (310928 + 5028046) = 5338974 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (310928 + 0 + 3287855 + 0) = 3598784 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (310928 + 3287855 + 0) = 3598784 N

Summary of Failure Path Calculations:

Path 1-1 = 5338974 N	, must exceed W = 2633211 N	or W1 = 3644958 N
Path 2-2 = 3598784 N	, must exceed W = 2633211 N	or W2 = 9328756 N
Path 3-3 = 3598784 N	, must exceed W = 2633211 N	or W3 = 9328756 N

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 6.43 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 216.7625 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1299.1005 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N1400 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	4364.000	mm
Vessel Thickness	Tv	73.000	mm
Design Temperature	T1	400.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	101.03	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	608.000	mm
Nozzle Thickness	Tn	120.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	101.00	MPa
Design Internal Pressure	Dp	6.411	MPa
Include Pressure Thrust		No	

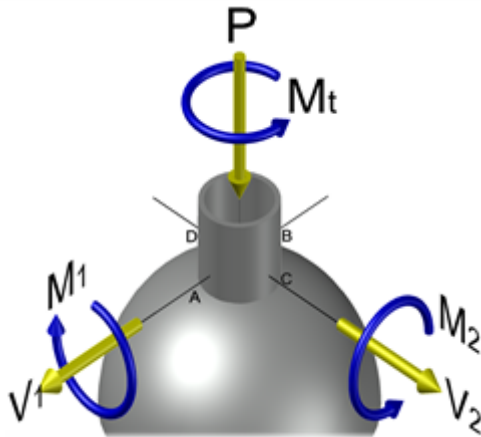
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	36000.0	N
Longitudinal Shear (SUS)	(V1) V1	26000.0	N
Circumferential Shear (SUS)	(Vc) V2	26000.0	N
Circumferential Moment (SUS)	(Mc) M1	32000000.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	32000000.0	N-mm
Torsional Moment (SUS)	Mt	36799944.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 608.0 + 2 \cdot 1.65 \cdot \sqrt{(2220.0 (73.0 - 3.0))}$$

$$= 1908.887 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	36000.0	N
Circumferential Shear	(VC) V2	26000.0	N
Longitudinal Shear	(VL) V1	26000.0	N
Circumferential Moment	(MC) M1	32000000.0	N-mm
Longitudinal Moment	(ML) M2	32000000.0	N-mm
Torsional Moment	MT	36799944.0	N-mm

Dimensionless Param: U = 0.77 TAU = 5.00 (2.10) RHO = 0.60

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.05183	(A,B,C,D)
$M(x) / P$	SP 1	0.05737	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.07100	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.11898	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.07100	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.11898	(A,B,C,D)
$N(y) * T / P$	SP 1	0.08959	(A,B,C,D)
$M(y) / P$	SP 1	0.01795	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.08609	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04352	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.08609	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04352	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4
Rad. Bend.	P	-2.5	2.5	-2.5	2.5	-2.5	2.5	-2.5	2.5
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-11.8	11.8	11.8	-11.8
Rad. Memb.	ML	-1.2	-1.2	1.2	1.2	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-11.8	11.8	11.8	-11.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-15.9	12.8	10.1	-8.5	-15.9	12.8	10.1	-8.5



Tang. Memb. P	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Tang. Bend. P	-0.8	0.8	-0.8	0.8	-0.8	0.8	-0.8	0.8
Tang. Memb. MC	0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Tang. Bend. MC	0.0	0.0	0.0	0.0	-4.3	4.3	4.3	-4.3
Tang. Memb. ML	-1.4	-1.4	1.4	1.4	0.0	0.0	0.0	0.0
Tang. Bend. ML	-4.3	4.3	4.3	-4.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-7.2	3.0	4.3	-2.8	-7.2	3.0	4.3	-2.8
Shear VC	0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tot. Shear	1.3	1.3	0.5	0.5	0.5	0.5	1.3	1.3
Str. Int.	16.1	13.0	10.1	8.5	15.9	12.8	10.4	8.8

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		100.1	100.1	100.1	100.1	100.1	100.1	100.1	100.1
Rad. Pl (SUS)		-1.6	-1.6	0.8	0.8	-1.6	-1.6	0.8	0.8
Rad. Q (SUS)		-14.4	14.4	9.3	-9.3	-14.4	14.4	9.3	-9.3
Long. Pm (SUS)		100.1	100.1	100.1	100.1	100.1	100.1	100.1	100.1
Long. Pl (SUS)		-2.1	-2.1	0.8	0.8	-2.1	-2.1	0.8	0.8
Long. Q (SUS)		-5.1	5.1	3.5	-3.5	-5.1	5.1	3.5	-3.5
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear Q (SUS)		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Pm (SUS)		100.1	100.1	100.1	100.1	100.1	100.1	100.1	100.1
Pm+Pl (SUS)		98.8	98.8	101.3	101.3	98.8	98.8	101.3	101.3
Pm+Pl+Q (Total)		93.1	113.1	110.3	97.4	93.0	112.9	110.5	97.6

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	100.12	101.03	Passed
Pm+Pl (SUS)	101.30	151.54	Passed
Pm+Pl+Q (TOTAL)	113.09	303.08	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: